UNIT V

CLOUD PLATFORMS AND APPLICATIONS

Cloud platform: Microsoft Azure, Amazon Web Services, Google Cloud, IBM Cloud and Cloud Linux–APIs - Microservices - Docker - Kubernetes - Applications and Use Cases.

Cloud platform: Microsoft Azure

Designed by Microsoft in 2010, Microsoft Azure is one of the widely used cloud computing platforms. Azure provides a wide variety of services, such as cloud storage, compute services, network services, cognitive services, databases, analytics, and IoT. It makes building, deploying, and managing applications very easy. All the Microsoft Azure fundamentals are also described to better understand readers.

Microsoft Azure tutorial is designed for beginners to learn about Microsoft Azure and its various cloud computing services. It covers important concepts and provides a comprehensive understanding of Azure's services. The tutorial includes a variety of topics that delve into the uses and services of Azure in depth.

Prerequisites to Learn Azure

To learn Microsoft Azure there is no need of any extra knowledge any one can easily learn the Azure if they have the below listed set of skills.

- Basic understanding of Azure concept.
- Understanding of Cloud Concepts
- Understanding if basic infrastructure management, database management, and software development.

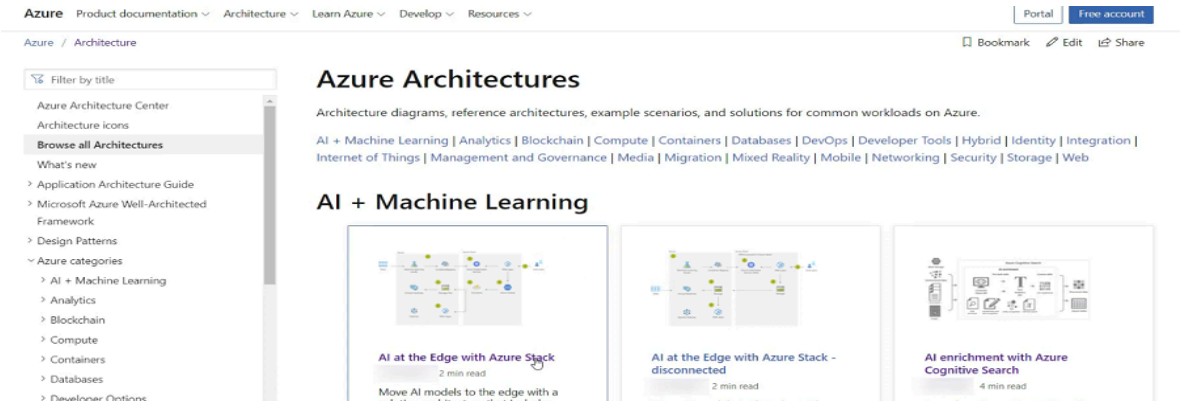
Introduction of Microsoft Azure

Discover how Azure empowers businesses to build, deploy, and manage applications across Microsoft's global network of data centers, supporting a wide array of programming languages, frameworks, and tools. Whether you're new to cloud computing or looking to expand your expertise, Azure provides the tools and scalability needed to meet diverse business needs effectively.

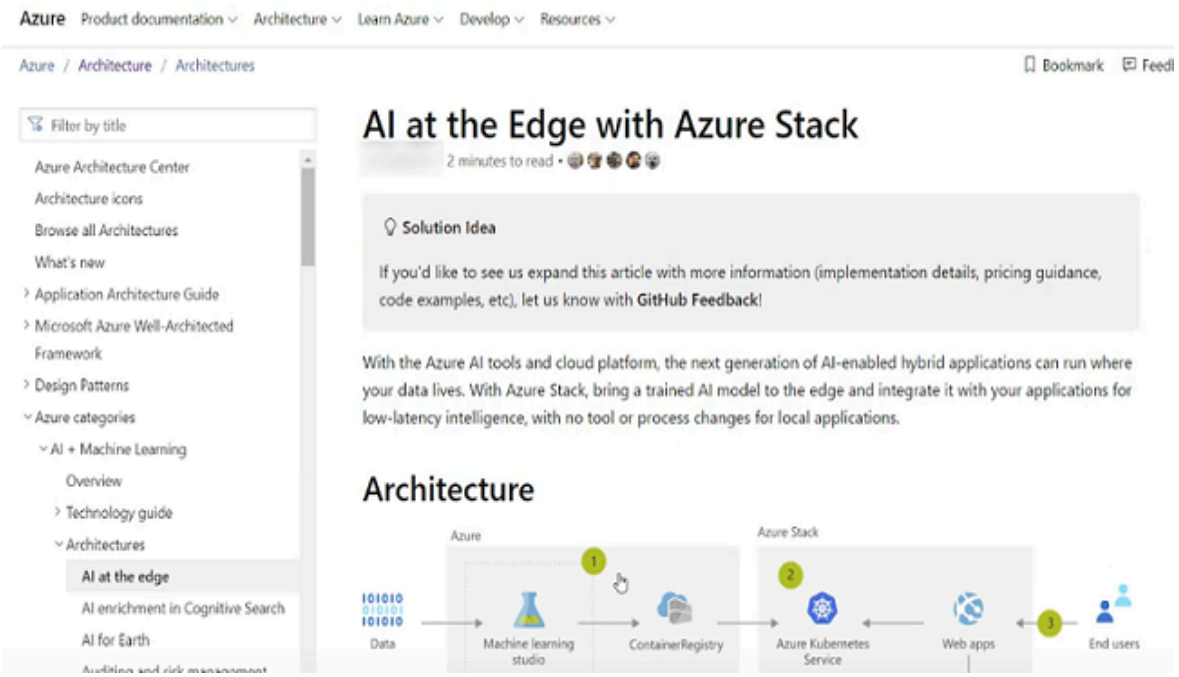
Microsoft Azure – High-Level Overview of Architecture Center:

All things related to the architecture in Azure can be found in the Azure Architecture Center. The Azure Architecture Center provides guides to design structured applications on Azure that are highly secure, scalable, and resilient. It is a collection of resources on Microsoft Docs. In this article, we will explore the same and get an overview of the Azure Architecture Center

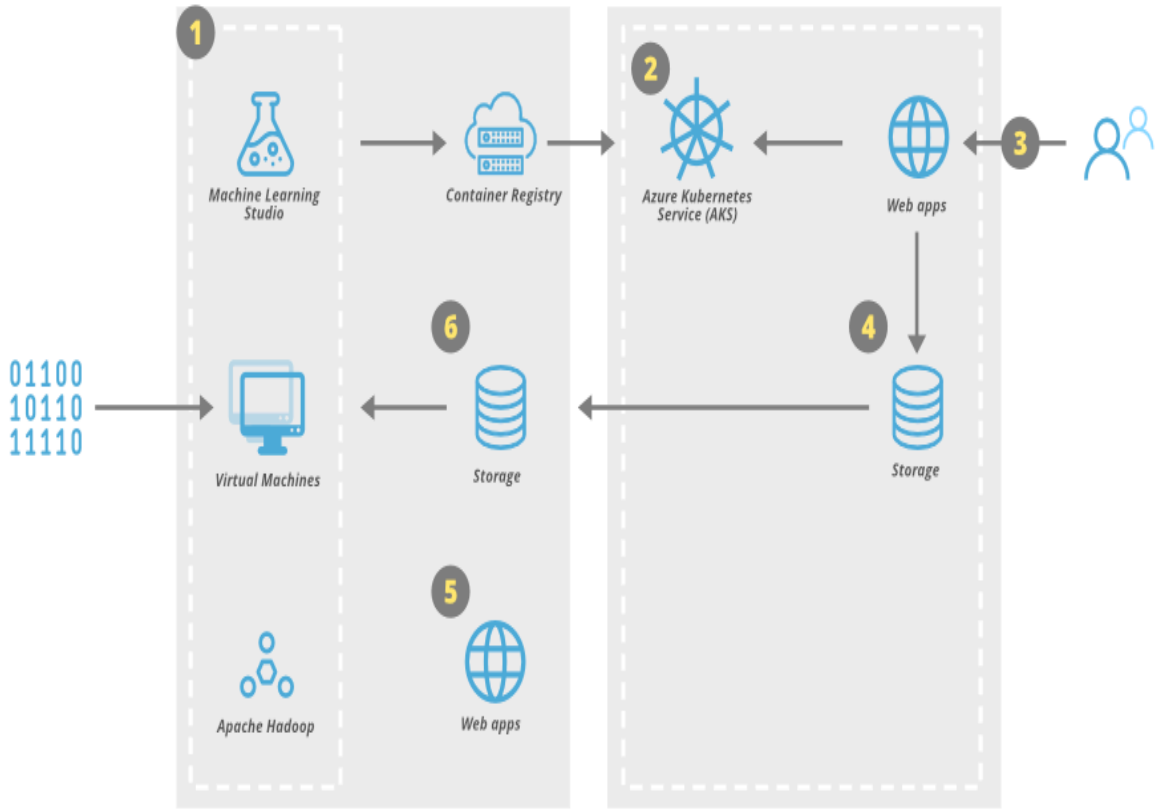
In the Azure Architecture Center, we can also browse more articles and many sample architectures. Let's browse some Azure architectures. To view available Azure Architectures click on the Browse all architecture in the left bar as shown below. From here, we can see many sample architectures for many scenarios as shown below:



For the purpose of example, we will check on the AI at the Edge with Azure. This takes us to the architecture for AI at the edge in Azure Stack and tells us about how data would flow and which components it contains.



For instance, it would show you that the Architecture for **AI at the Edge with Azure Stack Hub** as below:



It gives you standard design pattern and data flow information on AI at the Edge Azure Stack Hub like shown below:

Data Flow

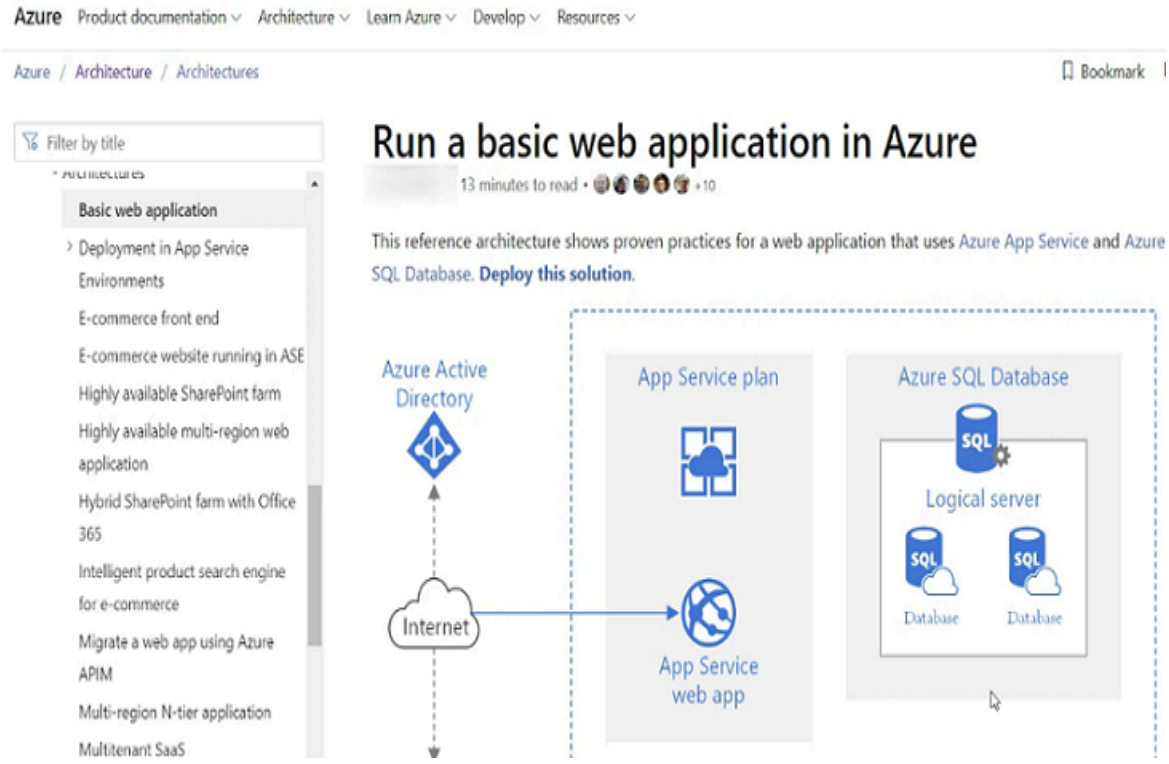
1. Data scientists train a model using Azure Machine Learning Studio (classic) and an HDInsight cluster. The model is containerized and put in to an Azure Container Registry.
2. The model is deployed via steps not represented in the diagram to a Kubernetes cluster on Azure Stack Hub.
3. End users provide data that is scored against the model.
4. Insights and anomalies from scoring are placed into storage for later upload.
5. Globally-relevant and compliant insights are available in the global app.
6. Data from edge scoring is used to improve the model.

It will also give you information regarding the various components working in conjunction to build the application as shown below:

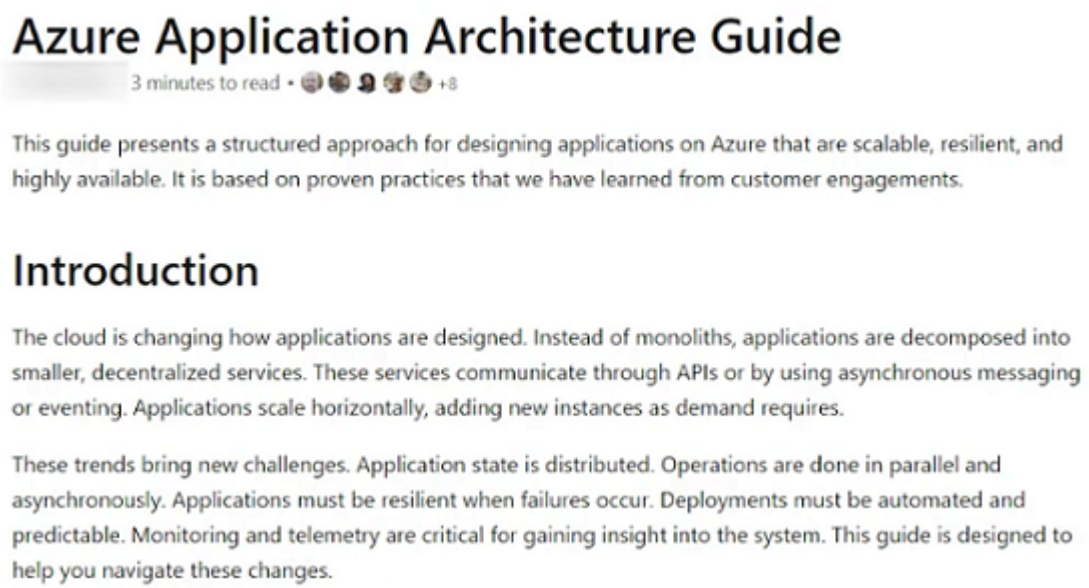
Components

- HDInsight: Provision cloud Hadoop, Spark, R Server, HBase, and Storm clusters
- Azure Machine Learning Studio (classic): Easily build, deploy, and manage predictive analytics solutions
- Virtual Machines: Provision Windows and Linux virtual machines in seconds
- Azure Kubernetes Service (AKS): Simplify the deployment, management, and operations of Kubernetes
- Storage: Durable, highly available, and massively scalable cloud storage
- Azure Stack Hub: Build and run innovative hybrid applications across cloud boundaries

There are so many sample architectures that vary in domains ranging from web-development, AI, machine learning, IoT, etc. Let's look into one named "**architectures for web applications"**. This contains detailed guidance and recommendations. It even lets you deploy the sample to Azure.



The architecture center also contains icons. You can download these for free and use them in your architectural designs. There's also the application architecture guide. This is a complete guide that tells you how to design an application for the Cloud. It is really comprehensive.



Azure Application Architecture Guide

3 minutes to read • +8

This guide presents a structured approach for designing applications on Azure that are scalable, resilient, and highly available. It is based on proven practices that we have learned from customer engagements.

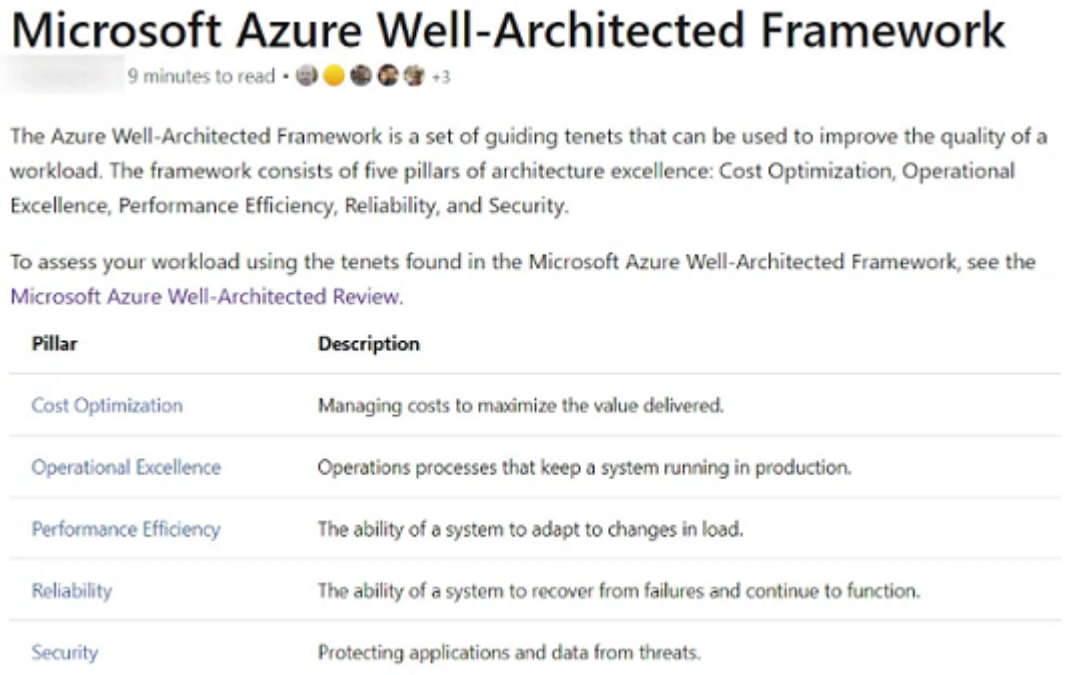
Introduction

The cloud is changing how applications are designed. Instead of monoliths, applications are decomposed into smaller, decentralized services. These services communicate through APIs or by using asynchronous messaging or eventing. Applications scale horizontally, adding new instances as demand requires.

These trends bring new challenges. Application state is distributed. Operations are done in parallel and asynchronously. Applications must be resilient when failures occur. Deployments must be automated and predictable. Monitoring and telemetry are critical for gaining insight into the system. This guide is designed to help you navigate these changes.

On top of that is the **Azure well-architectured framework**. You can use this framework to improve and optimize a workload. The framework takes you through the pillars of the following:

- Cost optimization
- Operational excellence
- Performance efficiency
- Reliability
- Security



Microsoft Azure Well-Architected Framework

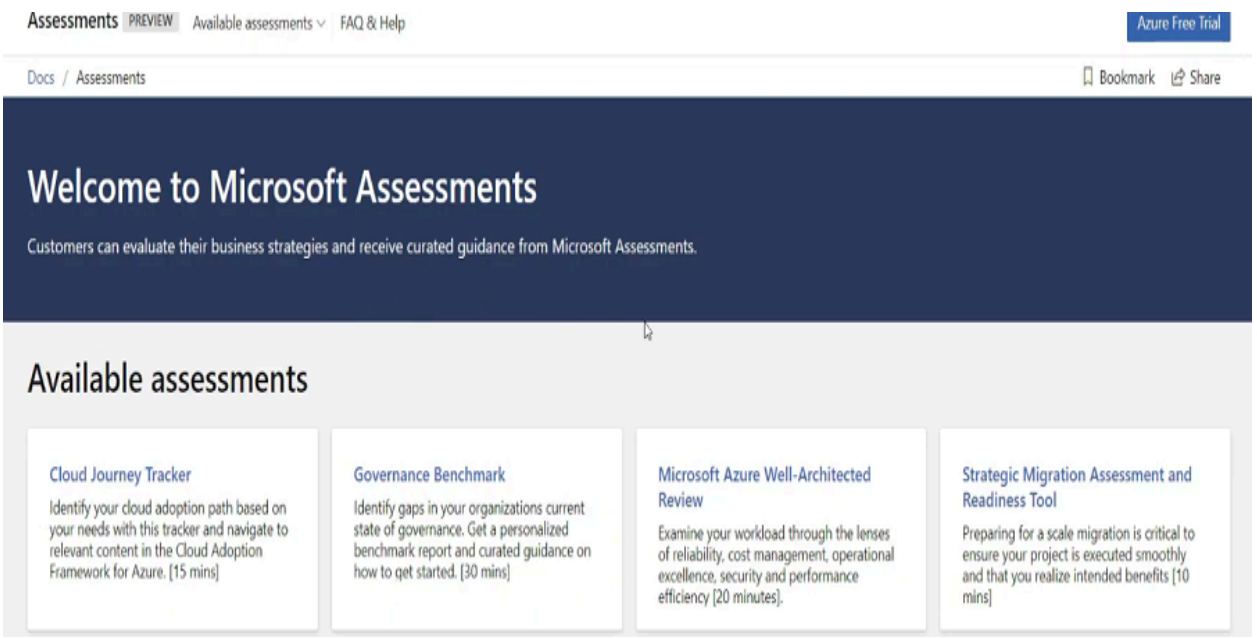
9 minutes to read • +3

The Azure Well-Architected Framework is a set of guiding tenets that can be used to improve the quality of a workload. The framework consists of five pillars of architecture excellence: Cost Optimization, Operational Excellence, Performance Efficiency, Reliability, and Security.

To assess your workload using the tenets found in the Microsoft Azure Well-Architected Framework, see the Microsoft Azure Well-Architected Review.

Pillar	Description
Cost Optimization	Managing costs to maximize the value delivered.
Operational Excellence	Operations processes that keep a system running in production.
Performance Efficiency	The ability of a system to adapt to changes in load.
Reliability	The ability of a system to recover from failures and continue to function.
Security	Protecting applications and data from threats.

The center also contains **Design Patterns** for the Cloud. These are solutions to well-known problems. You can use them as per your needs. Finally, you can use **Microsoft Assessments** to assess your architecture.



You can create an assessment that caters to your needs and you can take it by answering questions about your design. The Azure Architecture Center is an incredible resource for architecture guidance.

Azure is Microsoft's cloud computing platform which helps to build solutions to meet business goals. It supports infrastructure (IaaS), platform (PaaS), and software as a service (SaaS) computing services. It also supports advanced computing services like artificial intelligence, machine learning, and IoT. Azure allows you to build, manage and deploy the application on a global network.

How does Azure work:

It is a private and public cloud platform that helps developers and IT professionals to build deploy and manage the application. It uses the technology known as virtualization. Virtualization separates the tight coupling between the hardware and the operating system using an abstraction layer called a **hypervisor**. Hypervisor emulates all the functions of a computer in the virtual machine, it can run multiple virtual machines at the same time and each virtual machine can run any operating system such as Windows or Linux. Azure takes this virtualization technique and repeats it on a massive scale in Datacenter owned by Microsoft. Each data center has many racks filled with servers and each server includes the hypervisor to run multiple virtual machines. The network switch provides connectivity to all those servers.

One server in each rack runs a special piece of software called fabric controller. Each fabric controller is connected to another special piece of software known as the Orchestrator. Each Orchestrator is responsible for managing the work like responding to user requests.

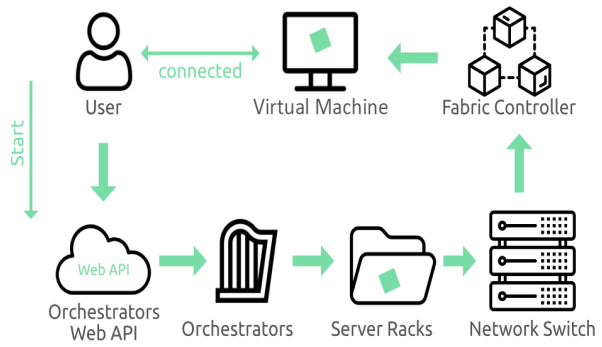
Microsoft Azure Services:

The following are the services that **Microsoft Azure offers:**

1. **Compute:** It includes Virtual Machines, Virtual Machine Scale Sets, Functions for serverless computing, Batch for containerized batch workloads, Service Fabric for microservices and container orchestration, and Cloud Services for building cloud-based apps and APIs.
2. **Networking:** With Azure, you can use a variety of networking tools, like the Virtual Network, which can connect to on-premise data centers; Load Balancer; Application Gateway; VPN Gateway; Azure DNS for domain hosting, Content Delivery Network, Traffic Manager, ExpressRoute dedicated private network fiber connections; and Network Watcher monitoring and diagnostics.
3. **Storage:** Includes Blob, Queue, File, and Disk Storage, as well as a Data Lake Store, Backup, and Site Recovery, among others.
4. **Web + Mobile:** Creating Web + Mobile applications is very easy as it includes several services for building and deploying applications.
5. **Containers:** Azure has a property that includes Container Service, which supports Kubernetes, DC/OS or Docker Swarm, and Container Registry, as well as tools for microservices.
6. **Databases:** Azure has also included several SQL-based databases and related tools.
7. **Data + Analytics:** Azure has some big data tools like HDInsight for Hadoop Spark, R Server, HBase, and Storm clusters.
8. **AI + Cognitive Services:** With Azure developing applications with artificial intelligence capabilities, like the Computer Vision API, Face API, Bing Web Search, Video Indexer, Language Understanding Intelligent.
9. **Internet of Things:** Includes IoT Hub and IoT Edge services that can be combined with a variety of machine learning, analytics, and communications services.
10. **Security + Identity:** Includes Security Center, Azure Active Directory, Key Vault, and Multi-Factor Authentication Services.
11. **Developer Tools:** Includes cloud development services like Visual Studio Team Services, Azure DevTest Labs, HockeyApp mobile app deployment and monitoring, Xamarin cross-platform mobile development, and more.

Creation And Connection of User and Virtual Machine:

Follow the below steps to create and connect a user to a virtual machine:



Step 1: The user makes a request using Orchestrators Web API for creating the virtual machine.
Step 2: The web API can be called by many tools including the UI of the Azure portal.
Step 3: The orchestrator package everything that is needed, it picks the best server rack and sends the package and request to the fabric controller.
Step 4: Once the fabric controller has created the virtual machine the user can connect to it.

Advantages of Azure:

Following are some **advantages of using Microsoft Azure:**

1. **High Availability:** It refers to the quality of computing infrastructure which allows it to continue functioning, even when some of its components fail.
2. **Data Security:** Azure provides many of the things to secure data over the cloud-like Microsoft Defender for Cloud, Key Vault, Azure Information Protection, and many more**.**
3. **Scalability:** Azure provides 2 types of scalability i.e. Vertical and Horizontal scaling to tackle the load by changing the capacity of resources or by adding the resources.
4. **Cost-Effective:** Azure provides different pricing models that can help to save costs.
5. **Learning-Curve:** Azure provides various programming languages such as C#, Visual Basics etc., and tools such as Visual Studio, Azure ML Studio, Azure Dev tools etc., for learning.
6. **Hybrid-Capabilities:** Azure provides hybrid working model. It allows the organization or enterprise to avail services from public cloud as well as from on-premise network.

Disadvantages of Azure:

Following are the key disadvantages of using Microsoft Azure:

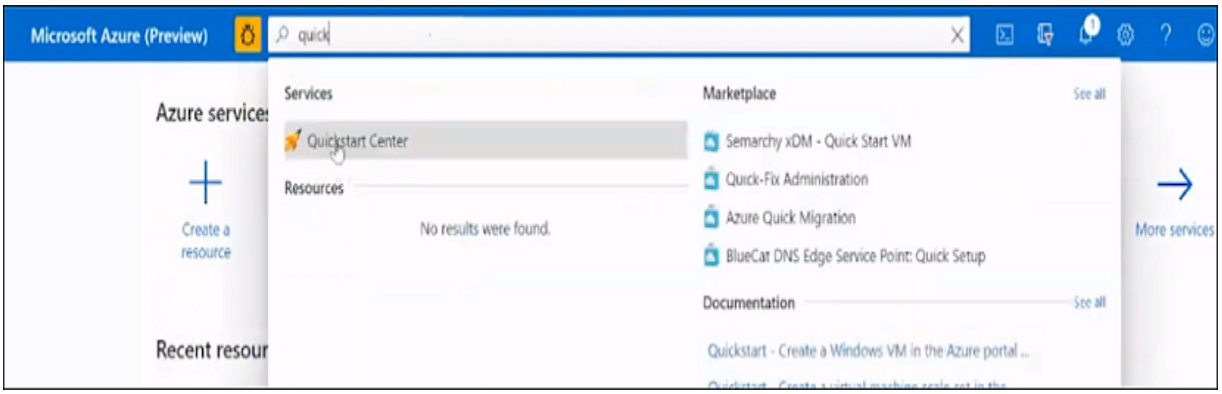
1. **Requires Platform Expertise:** A common mistake, on-premise servers compute power does not translate equivocally in the cloud but can potentially cost businesses.
2. **Requires Management:** As Azure provides many of the platforms so it needs to be expertly managed and maintained which includes patching and server monitoring.
3. **Complexity:** For maintaining the SaaS application for large enterprise, it is bit complexity in nature to ensure the smoothness of application all the time.

Microsoft Azure – Using the Azure Quickstart Centre

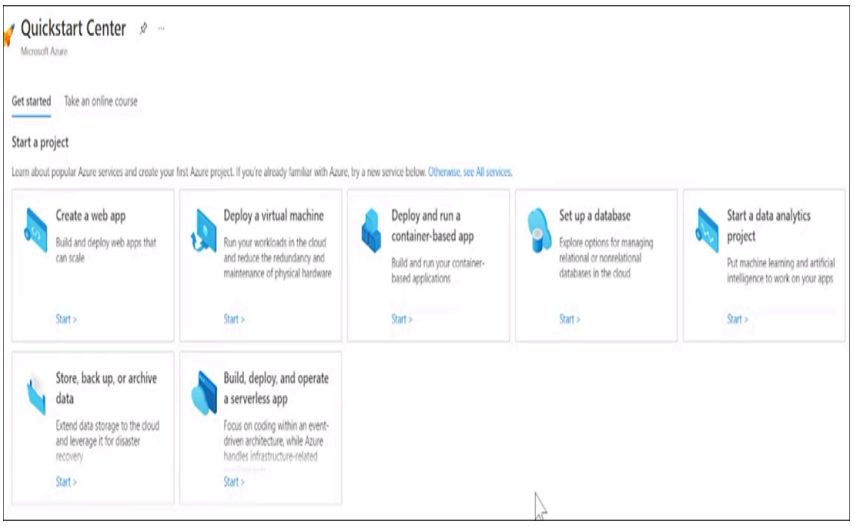
In this article, we will learn how to get started easily with Azure using Azure Quickstart Center. The Azure Quickstart Center is a tool in the Azure Portal to help you find the right services for you to get started in Azure with confidence. A little over a year ago Azure made the Quickstart Center generally available, and it offers you three main ways to get started.

Implementation:

So, how do we find the Quickstart Center? The most helpful way is just to type it into the search and you can see that there's a little rocket icon. So, here we are in the Quickstart Center.

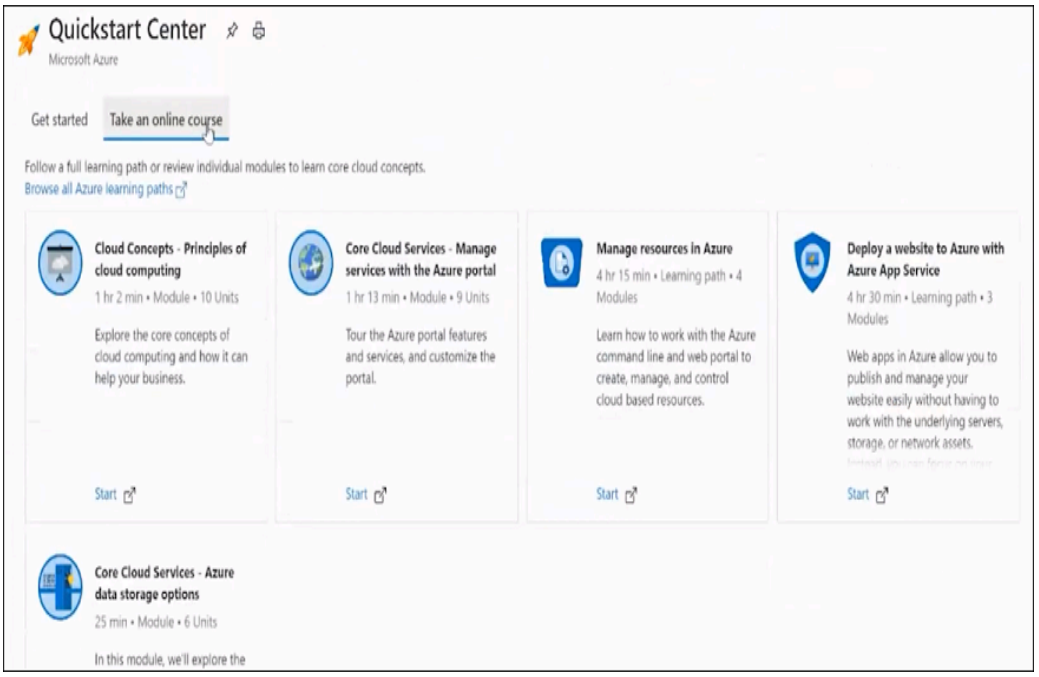


Here, you will see, there are three main ways to get started. First, you can choose from some of these popular scenarios in order to start the first project. In this article, we are going to look into how to find the right service for you in order to start your first project. So, you have a couple of different scenarios that you can choose from. You can create a web app, deploy a virtual machine, started a data analytics project, or back up some of your data.

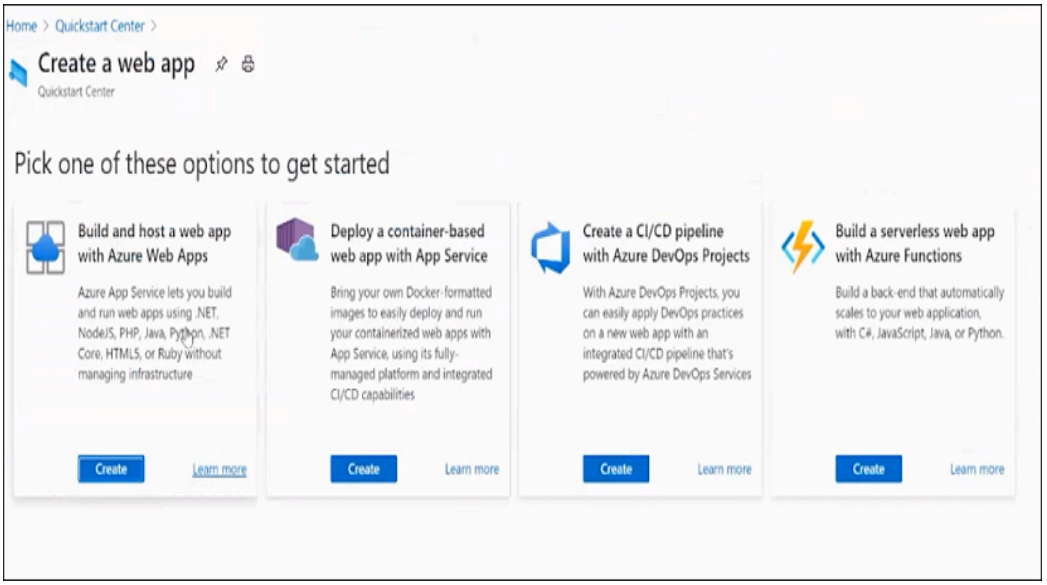


You can look through some of the guides that help you to understand how to set up your Azure environment, how to migrate to the cloud from the on-premise environment, or how to drive innovation in your cloud-first applications.

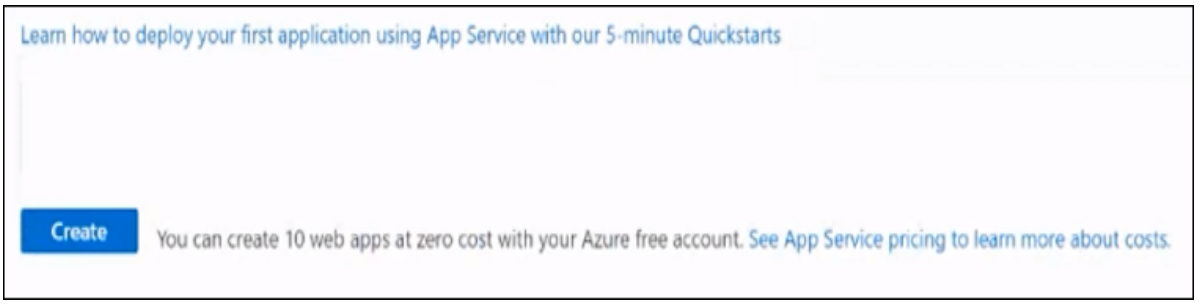
You can also take an online course if you want to learn more about some of the Azure Fundamental concepts.



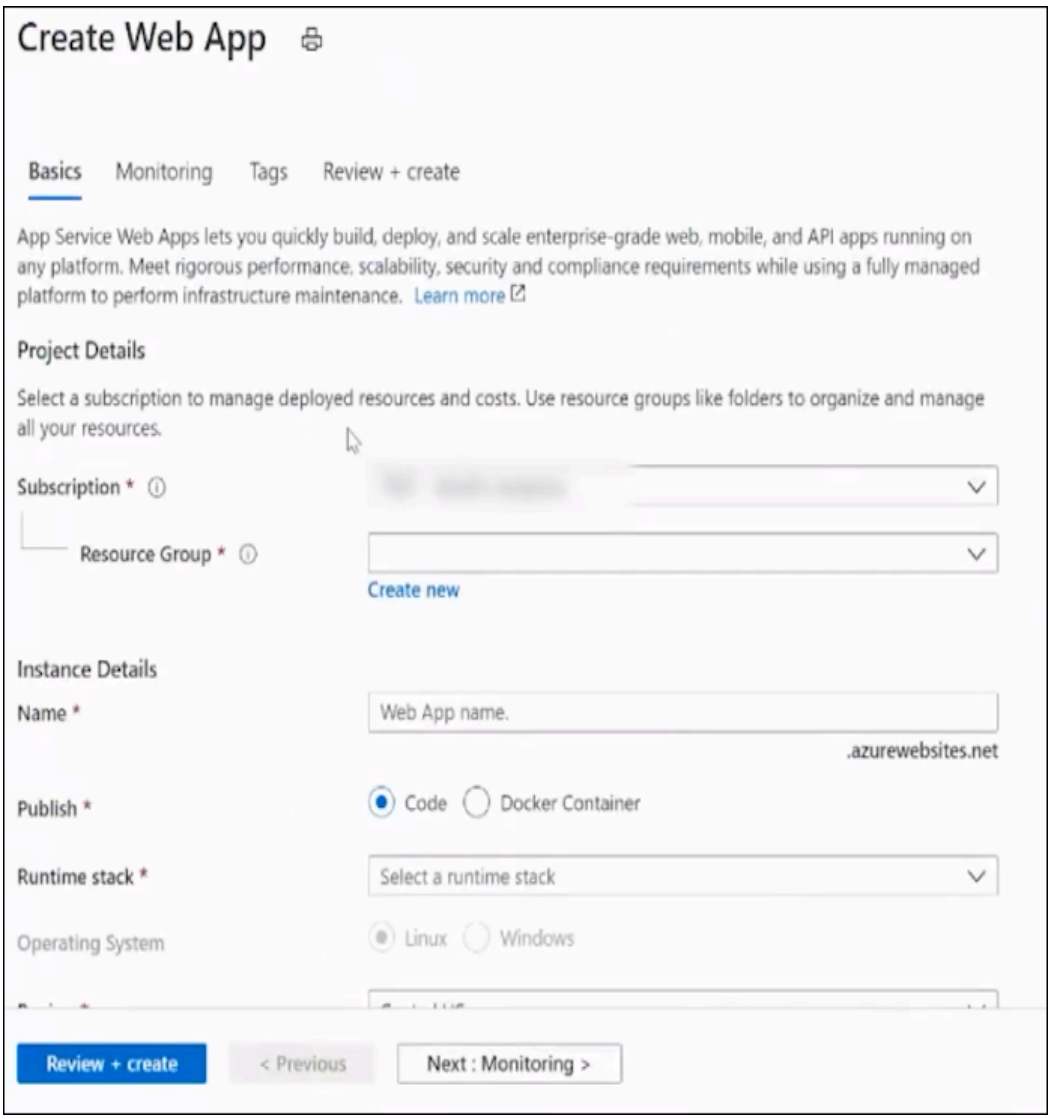
Let's create a web app. So, in order to do so, you have to go to click here and you can see, there are a couple of different ways that you can get started to create a web app in Azure. Azure offers a couple of different services. So, you can use Azure Web Apps, container-based web apps. I can do it through DevOps or I can even create a serverless web app through Azure. Functions.



To learn the differences between these, you can click on the card where you can learn more and understand what is this in the first place? Why would I want to use it? How does this web app connect to other services in Azure through this reference architecture diagram? What are some of the prerequisites that I might need and see about pricing? And you can see here that you can create ten web apps for free with my Azure free account.



Here, click on create. You will be landed on the create a form for the web app and you just need to fill out a few key details before you can create my web app resource. So, you just need to fill out the subscription, resource group, or you can create a new one, the name of your web app, pick your runtime stack and operating system as well as region. And, just click review and create.



Amazon Web Services (AWS)

Amazon Web Services (AWS) is a cloud computing platform offered by Amazon. It provides a wide range of on-demand services like computing power, storage, and databases, allowing businesses to scale and manage their IT resources efficiently. AWS offers services such as EC2 for virtual servers, S3 for scalable storage, RDS for managed databases, and Lambda for serverless computing. By using AWS, companies can reduce infrastructure costs, improve flexibility, and deploy applications globally with ease.

Amazon Web Services (AWS) is a leading top platform in providing the web services of various domains. AWS follows the trends of digital IT and comes up needy services with optimized performances covering a wide range of services from Compute to Storage. It covers a wider range of customers of different domains to expand their business operations. This Article covers the fundamentals of AWS and its scope of IT business.

What Is AWS And Why Is It Used?

AWS stands for Amazon Web Services, It is an expanded **cloud computing platform** provided by Amazon Company. AWS provides a wide range of services with a **pay-as-per-use pricing model** over the Internet such as Storage, Computing power, Databases, Machine Learning services, and much more. AWS facilitates for both businesses and individual users with effectively hosting the applications, storing the data securely, and making use of a wide variety of tools and services improving management flexibility for IT resources.

History Of AWS

Then providing Simple Storage Service (Amazon S3) revolutionized with scalable management of Storage. Coming up with effective compute and storage services and providing them rental basis helped many startup companies and users with the cost of manual Hardware Infrasture setup. Introducing the concept of serverless computing with AWS lambda services enhanced its business globally. It came up with services like Elastic Beanstalk made the deployment of applications much easier bringing large audiences. AWS always came with diverse array of services offering with technical innovations, updated services with current trends. AWS has emerged as a powerhouse in the world of Cloud Computing.

AWS Works?

AWS comes up with its own network infrastructure on establishing the datacenters in different regions mostly all over the world. Its global Infrastructure acts as a backbone for operations and services provided by AWS. It facilitates the users on creating secure environments using Amazon VPCs (Virtual Private Clouds). Essential services like Amazon EC2 and Amazon S3 for utilizing the compute and storage service with elastic scaling. It supports the dynamic scaling of the applications with the services such as Auto Scaling and Elastic Load Balancing (AWS ELB). It provides a good user-friendly AWS Management Console facilitating seamless configuration and management of AWS services to the Users. Its Architecture ensures high availability , fault tolerance making AWS as a versatile powerful Cloud Computing Platform.

AWS Fundamentals

In the Journey of AWS, understanding the key concepts such as Regions, Availability Zones, Global Network Infrastructure, etc is crucial. The fundamentals of AWS keep on maintaining the applications reliable and scalable with services globally with coming to a strategic deployment of resources for optimal performance and resilience. The following are the some of the main fundamentals of AWS:

- **Regions:** AWS provide the services with respective division of regions. The regions are divided based on geographical areas/locations and will establish data centers. Based on need and traffic of users, the scale of data centers is depended to facilitate users with low-latencies of servcies.

- **Availability Zones (AZ):** To prevent the Data centers for the Natural Calamities or any other disasters. The Datacenters are established as sub sections with isolated locations to enhance fault tolerance and disaster recovery management.
- **Global Network Infrastructure:** AWS ensures the reliability and scalability of services through setting up its own AWS Network Infrastructure globally. It helps in better management of data transmissions for optimized performance and security reliance.

Top AWS Services

In the rapid revolution of Cloud Computing, AWS facilitates with wide variety of services respect to the fields and needs. The following are the top AWS services that are in wide usage:

- **Amazon EC2(Elastic Compute Cloud) :** It provides the Scalable computing power via cloud allowing the users to run applications and manage the workloads over their remotely.
- **Amazon S3 (Simple Storage Service):** It offers scalable object Storage as a Service with high durability for storing and retrieving any amount of data.
- **AWS Lambda**: It is a service in Serverless Architecture with Function as a Service facilitating serverless computing i.e., running the code on response to the events, the background environment management of servers is handled by aws automatically. It helps the developers to completely focus on the logic of code build.
- **Amazon RDS (Relational Database Service):** This is an aws service that simplifies the management of database providing high available relational databases in the cloud.
- **Amazon VPC (Virtual Private Cloud):** It enables the users to create isolated networks with option of public and private expose within the AWS cloud, providing safe and adaptable configurations of their resources.

To know more about refer the Article – Top 25 AWS Service List

Advantages Of Amazon Web Services

- AWS allows you to easily scale your resources up or down as your needs change, helping you to save money and ensure that your application always has the resources it needs.
- AWS provides a highly reliable and secure infrastructure, with multiple data centers and a commitment to 99.99% availability for many of its services.
- AWS offers a wide range of services and tools that can be easily combined to build and deploy a variety of applications, making it highly flexible.
- AWS offers a pay-as-you-go pricing model, allowing you to only pay for the resources you actually use and avoid upfront costs and long-term commitments.

Disadvantages Of Amazon Web Services

- AWS can be complex, with a wide range of services and features that may be difficult to understand and use, especially for new users.
- AWS can be expensive, especially if you have a high-traffic application or need to run multiple services. Additionally, the cost of services can increase over time, so you need to regularly monitor your spending.

- While AWS provides many security features and tools, securing your resources on AWS can still be challenging, and you may need to implement additional security measures to meet your specific requirements.
- AWS manages many aspects of the infrastructure, which can limit your control over certain parts of your application and environment.

Applications Of AWS

The AWS services are using by both startup and MNC companies as per their usecase. The startup companies are using overcome hardware infrasture cost and applications deployments effectively with cost and performance. Whereas large scale companies are using AWS cloud services for the management of their Infrastructure to completely focus on the development of products widely. The following the Real-world industrial use-cases of AWS services:

- **Netflix:** The Large streaming gaint using AWS for the storage and scaing of the applications for ensuring seamless content delivery with low latency without interruptions to millions of users globally.
- **Airbnb:** By utilizing AWS, Airbnb manages the various workloads and provides insurable and expandable infrastructure for its virtual marketplace and lodging offerings.
- **NASA's Jet Propulsion Laboratory:** It takes the help of AWS services to handle and analyze large-scale volumes of data related to vital scientific research missions and space exploration.
- **Capital One:** A financial Company that is utilizing AWS for its security and compliance while delivering innovative banking services to its customers.

AWS Global Infrastructure

The AWS global infrastructure is massive and is divided into geographical regions. The geographical regions are then divided into separate availability zones. While selecting the geographical regions for AWS, three factors come into play

- Optimizing Latency
- Reducing cost
- Government regulations (Some services are not available for some regions)

Each region is divided into at least two availability zones that are physically isolated from each other, which provides business continuity for the infrastructure as in a distributed system. If one zone fails to function, the infrastructure in other availability zones remains operational. The largest region North Virginia (US-East), has six availability zones. These availability zones are connected by high-speed fiber-optic networking.

There are over 100 edge locations distributed all over the globe that are used for the CloudFront (content delivery network). CloudFront can cache frequently used content such as images and videos(live streaming videos also) at edge locations and distribute it to edge locations across the globe for high-speed delivery and low latency for end-users. It also protects from DDOS attacks.

AWS Management Console

The AWS management console is a web-based interface to access AWS. It requires an AWS account and also has a smartphone application for the same purpose. So When you sign in

for first time, you see the console home page where you see all the services provided by AWS. Cost monitoring is also done through the console.

AWS resources can also be accessed through various Software Development Kits (SDKs), which allows the developers to create applications as AWS as its backend. There are SDKs for all the major languages(e.g., JavaScript, Python, Node.js, .Net, PHP, Ruby, Go, C++). There are mobile SDKs for Android, iOS, React Native, Unity, and Xamarin. AWS can also be accessed by making HTTP calls using the AWS-API. AWS also provides a AWS Command Line Interface (CLI) for remotely accessing the AWS and can implement scripts to automate many processes. This Console is also available as an app for Android and iOS. For mobile apps, you can simply download AWS console app.

AWS Cloud Computing Models

There are three cloud computing models available on AWS.

1. **Infrastructure as a Service (IaaS):** It is the basic building block of cloud IT. It generally provides access to data storage space, networking features, and computer hardware(virtual or dedicated hardware). It is highly flexible and gives management controls over the IT resources to the developer. For example, VPC, EC2, EBS.
2. **Platform as a Service (PaaS):** This is a type of service where AWS manages the underlying infrastructure (usually operating system and hardware). This helps the developer to be more efficient as they do not have to worry about undifferentiated heavy lifting required for running the applications such as capacity planning, software maintenance, resource procurement, patching, etc., and focus more on deployment and management of the applications. For example, RDS, EMR, ElasticSearch.
3. **Software as a Service(SaaS):** It is a complete product that usually runs on a browser. It primarily refers to end-user applications. It is run and managed by the service provider. The end-user only has to worry about the application of the software suitable to its needs. For example, Saleforce.com

What is Google Cloud Platform (GCP)

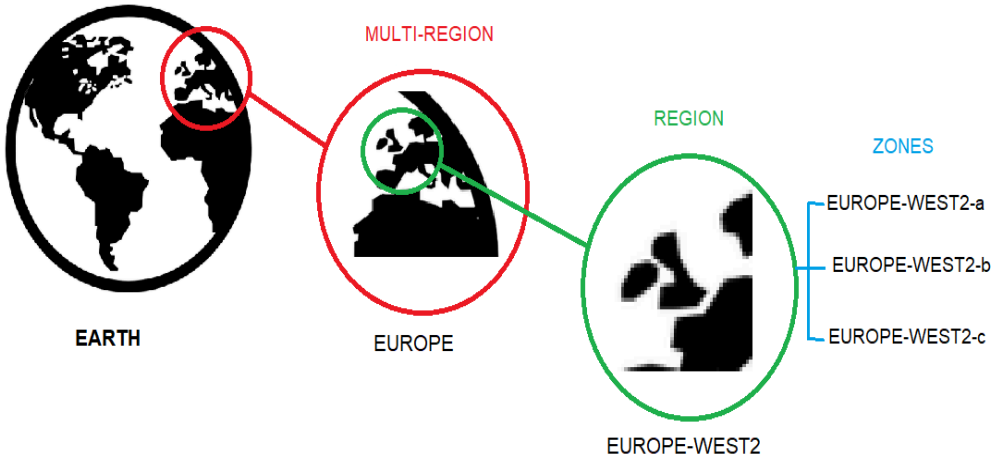
Google Cloud Platform (GCP) is a suite of cloud computing services provided by Google. It offers a wide range of services, including computing power, storage, databases, machine learning, networking, and more, all delivered over the internet. GCP enables businesses to build, deploy, and scale applications and services quickly and efficiently without the need to invest in or manage physical infrastructure.

Starting from 1998 with the launch of Google Search. Google has developed one of the largest and most powerful IT Infrastructures in the world. Today, this infrastructure is used by billions of users to use services such as Gmail, YouTube, Google Photos, and Maps. In 2008, Google decided to open its network and IT infrastructure to business customers, taking an infrastructure that was initially developed for consumers' applications to public service and launching the Google Cloud platform.

All the services listed above are provided by Google, hence the name Google Cloud Platform (GCP). Apart from these, there are many other services provided by GCP and many concepts related to them that we are going to discuss in this article.

Regions and Zones

Let's start at the finest grain level (i.e. the smallest or first step in the hierarchy), the **Zone**. A zone is an area where Google Cloud Platform Resources like virtual machines or storage are deployed.



For example, when you launch a virtual machine in GCP using Compute Engine, it runs in a zone you specify (suppose Europe-west2-a). Although people consider a zone as being sort of a GCP Data Center, that's not strictly accurate because a zone doesn't always correspond to one physical building. You can still visualize the zone that way, though.

Zones are grouped into **regions** which are independent geographic areas and much larger than zones (for example- all zones shown above are grouped into a single region Europe-west2) and you can choose what regions you want your GCP resources to be placed in. All the zones within a neighbourhood have fast network connectivity among them. Locations within regions usually have trip network latencies of under five milliseconds.

As a part of developing a fault-tolerant application, you'll need to spread your resources across multiple zones in a region. That helps protect against unexpected failures. You can run resources in different regions too. Lots of GCP customers do this, both to bring their applications closer to users around the world, and also to guard against the loss of a whole region, say, due to a natural disaster.

A few GCP Services supports deploying resources in what we call a **Multi-Region**. For example, Google Cloud Storage, lets you place data within the Europe Multi-Region. What that means is that it is stored redundantly in a minimum of two different geographic locations, separated by at least 160 kilometers within Europe. Previously, GCP had 15 regions. Visit cloud.google.com to ascertain what the entire is up to today.

History of Google Cloud Platform

Starting from 1998 with the launch of Google Search. google has developed one of the largest and most powerful IT Infrastructures in the world. Today, this infrastructure is used by billions of users to use services such as Gmail, YouTube, Google Photos, and Maps. In 2008, Google decided to open its network and IT infrastructure to business customers, taking an infrastructure that was initially developed for consumers' applications to public service and launching the Google Cloud platform. Over the next decade, Google expanded its offerings. Key milestones included the launch of BigQuery in 2010 for serverless analytics, Cloud Storage in 2013 and Compute Engine in 2014 offering Infrastructure-as-a-Service (IaaS). The debut of Google Kubernetes Engine (GKE) in the same year revolutionized container management setting GCP apart as a leader in cloud innovation. Today GCP is a powerhouse in cloud computing offering cutting-edge solutions that empower businesses to innovate, scale and succeed in a digital-first world.

Google Cloud Services Interaction

Google Cloud Platform (GCP) offers three primary methods for interacting with its services and resources:

1. Google Cloud Console

The Google Cloud Console is a web-based, graphical interface that allows you to manage and configure your GCP projects and resources. You can either create a new project or select an existing one to use resources within the project. The console provides an easy-to-navigate dashboard to monitor and control various Google Cloud services.

2. Command-Line Interface (CLI)

For those who prefer command-line operations, Google Cloud provides the Cloud SDK, which includes the gcloud CLI. This tool allows you to manage GCP resources directly from a terminal window. For example, to create a Compute Engine virtual machine (VM), you can use the `gcloud compute instances create` command. You can use the gcloud CLI in two ways:

- Install it locally on your computer.
- Use Cloud Shell, a browser-based terminal environment accessible directly from the Google Cloud Console, eliminating the need for local installation. Cloud Shell provides features such as a built-in code editor, 5 GB of persistent storage, and pre-installed tools, including the gcloud CLI. It supports multiple programming languages like Java, Python, Go, Node.js, and more.

To know more about GCP installation in windows & linux you can refer to this link How To Install GCP CLI In Linux, Windows ?

3. Client Libraries

Google Cloud also offers client libraries that simplify resource management and application development. These libraries expose APIs tailored to specific languages such as Python and Node.js, allowing you to interact with GCP services more intuitively. Client libraries are available for:

- App APIs for accessing services with less code and seamless integration with GCP.
- Admin APIs for managing resources, ideal for building automation tools.

Additionally, these libraries can be used for services like Google Maps, Drive, and YouTube.

Google Cloud vs Google Cloud Platform

Aspect	Google Cloud	Google Cloud Platform (GCP)
Definition	A broad suite of services offered by Google over the internet to support digital transformation.	A subset of Google Cloud that provides public cloud infrastructure for hosting applications.
Key Services	Includes products like Google Workspace, Android Enterprise, Chrome OS, and machine learning APIs.	Focused on cloud infrastructure services such as computing, storage, networking, and databases.
Example Offerings	• Google Workspace (Gmail, Docs, Drive, etc.) • Android and Chrome OS for enterprise • APIs for machine learning and mapping services	• Compute Engine • App Engine • Cloud Storage • Kubernetes Engine • BigQuery
Target Use Case	Provides collaboration tools, device management, and APIs for enterprise use.	Enables businesses to run and scale web-based applications on the cloud.

Google Cloud Platform Services

Here are some of the services offered by Google Cloud Platform categorized by their functionalities:

Category	Service	Description
Cloud Computing	Compute Engine	Compute Engine lets you create and manage virtual machines, giving you control over the resources you need, like memory, storage, and security settings for your applications.
	Google Kubernetes Engine (GKE)	GKE is a managed Kubernetes service that helps you deploy applications while automatically handling scaling and load balancing for you.
	App Engine	App Engine is a scalable platform for running web applications. It dynamically adjusts to changing demand and provides a secure environment to ensure your app performs well.

Category	Service	Description
Storage	Cloud Storage	Cloud Storage is designed to store large amounts of data that need to be highly available and easily accessed.
	Persistent Disk	Persistent Disk provides durable storage that can be attached to virtual machines and reused when necessary, making it versatile for various use cases.
	Cloud SQL	Cloud SQL is a fully managed database service that supports MySQL, PostgreSQL, and SQL Server, taking the hassle out of database management.
Networking	Virtual Private Cloud (VPC)	With VPC, you can run your applications inside a private network, offering more control and security for your infrastructure.
	Cloud Load Balancing	This service ensures your application traffic is distributed evenly across multiple instances, helping your app stay fast and responsive.
	Cloud CDN	Cloud CDN caches content and delivers it from the closest edge location to users, speeding up delivery and reducing load times.
Data Analytics	BigQuery	BigQuery is a powerful data warehouse that makes analyzing huge amounts of data easy and fast, giving organizations deep insights from their data.
	Dataflow	Dataflow helps you understand how data flows through your system, allowing you to optimize and analyze data processes for better performance.
	Pub/Sub	Pub/Sub is a messaging system that decouples services, allowing them to communicate asynchronously, improving system efficiency and preventing bottlenecks.
Machine Learning	Vertex AI Platform	Vertex AI Platform is Google Cloud's suite for building and managing AI models, helping organizations unlock the potential of AI for business transformation.

Category	Service	Description
	AI Platform Training	AI Platform Training lets you train machine learning models in the cloud, giving you the computing power to handle complex AI tasks.
	AI Platform Prediction	This service allows you to make predictions using your trained machine learning models, helping you apply AI to real-world problems.
Productivity and Collaboration	Google Workspace	Google Workspace includes popular tools like Gmail, Calendar, and Drive, helping teams collaborate seamlessly and stay organized in their daily work.
	Cloud Identity and Access Management (IAM)	IAM allows administrators to control who can access what within an organization, ensuring that only authorized users have the appropriate level of access to resources.

Must Read – *How To Create a Free Tier Account on GCP?*

Higher-Level Services on Google cloud

Here are some of the higher-level services offered by Google Cloud Platform:

- **Big Data and Analytics Services:** Big Data and Analytics Services offer insights from large volumes of data to help businesses make informed decisions.
- **Machine Learning and AI Services:** Machine learning and AI services are technologies that enable computers to learn from data and perform tasks without being explicitly programmed.
- **Serverless Computing:** Serverless computing is a cloud computing model where the cloud provider manages the infrastructure, allowing developers to focus solely on writing and deploying code without worrying about servers.

Use Cases of Google Cloud Platform

Google Cloud Platform is well suited for the build and deploy and manage the applications.

- **E-commerce:** You deploy and manage the e-commerce websites by autoscaling and load balancing you can manage the millions of users and transactions.
- **Media and entertainment:** You can store the static and dynamic data can deliver it to the across the world with out any latency to the end users.
- **Financial services:** Google Cloud Platform is well suited for the sinical application because of the level of security it is offering.
- **Healthcare:** You can store the data of patient and take care the outcomes of the health of patient.

Security in Google Cloud Platform

Google Cloud Platform offers following security options.

- **Encryption:** Google cloud platform offers security like encryption at rest and in transit for all of your data.
- **Access control:** You can control the access to the individual users like which services they can access and which service they can't depending on the use cases.
- **Network Security:** You can create the VPC where you can secure the application by deploying the application in the private network and also you can configure the firewalls and security groups etc.
- **Identity-Aware Proxy (IAP):** With IAP, users may manage application access according to their context and identity. It aids in preventing unwanted access.

Future of Google Cloud Platform

Google Cloud Platform (GCP) is evolving constantly by expanding its resources and increasing its regions and availability zone across the world which make it mote available for the users to use reduces the latency. GCP is upgrading itself according to the market trends GCP play an major role in the upcoming years it will play major role it will helps for the businesses to thrive in the increasingly data-driven and interconnected world.

- Artificial intelligence (AI) and Machine Learning (ML)
- Edge computing
- Data analytics and data management
- Cybersecurity
- Sustainability

GCP Pricing and Free Tier

Google was the primary major Cloud provider to **bill by the second** instead of rounding up to greater units of your time for its virtual machines as a service offering. This may not sound like a big deal, but charges for rounding up can really add up for customers who are creating and running lots of virtual machines. Per second billing is obtainable for a virtual machine use through Compute Engine and for several other services too.

Compute Engine provides automatically applied use discounts which are **discounts** that you simply get for running a virtual machine for a big portion of the billing month. When you run an instance for at least 25% of a month, Compute Engine automatically gives you a reduction for each incremental minute you employ it. Here's one more way Compute Engine saves you money.

Normally, you choose a virtual machine type from a typical set of those values, but Compute Engine also offers custom virtual machine types, in order that you'll fine-tune the sizes of the virtual machines you use. That way, you'll **tailor your pricing** for your workloads.

Free Tier: Google Cloud Platform offers a generous free tier with limited usage of various services, allowing users to explore and experiment with GCP without incurring charges. The free tier typically includes a certain amount of usage for services like Compute Engine, App Engine, Cloud Storage, BigQuery, and more.

IBM Cloud:

IBM Cloud (formerly known as IBM Bluemix and IBM SoftLayer) is a suite of cloud computing services that offers both Platform as a Service (PaaS) and Infrastructure as a Service (IaaS). It is currently one of the top five cloud service providers in the world.

At present, it holds a global market share of 3%. This puts IBM Cloud in fifth position worldwide, behind Google Cloud Platform (at 11%), Microsoft Azure (21%), Amazon Web Services (34%), and Alibaba Cloud (5% market share).

IBM Cloud has achieved this position through a combination of its own long-standing experience and reputation in the IT business, extensive links with organisations in a wide range of industries, and the provision of a comprehensive selection of cloud services, tools, and infrastructure.

IBM Cloud History

IBM acquired SoftLayer, a public cloud platform, in 2013, to serve as the foundation for its Infrastructure as a Service offering. IBM added Platform as a Service to the mix in 2016, when it rolled the SoftLayer brand under its own Bluemix brand. IBM rebranded its entire cloud portfolio as IBM Cloud in 2017.

Creating an IBM ID and Enabling Connection to IBM Cloud



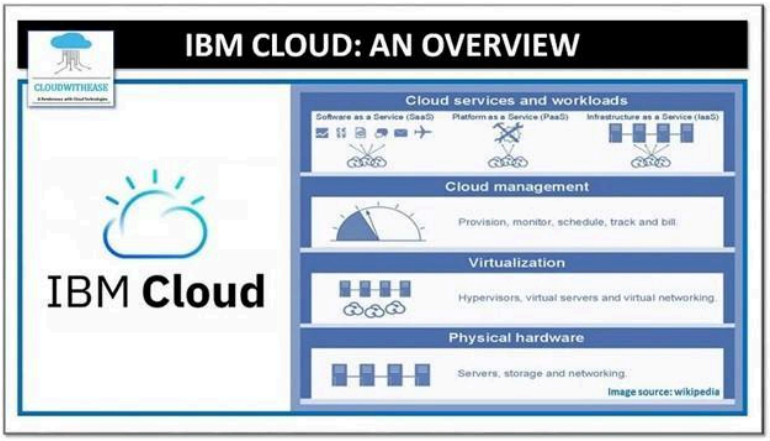
Image source: Cloudwithease]

To create an IBM Cloud account, you first need to go to the IBM Cloud Login website and enter your IBMid email address. If you don't have one of these already, an ID will be created based on the valid email address that you enter.

To enable a cloud connection to IBM Cloud, you can use the IBM management GUI or command-line interface. If a DNS server is not already configured on the system, enter the IP address and name of the server that converts the host name of the cloud service provider to IP addresses for the system.

Note that you cannot configure encryption after the IBM Cloud account is created. Before proceeding, you should therefore ensure that you understand the security considerations relating to cloud storage and encryption, when planning for transparent cloud tiering.

IBM Cloud Components and Resources



[Image source: Cloudwithease]

IBM Cloud's Infrastructure as a Service offerings enable organisations to deploy and access virtualized IT resources such as compute power, storage and networking over the internet. Subscribers can choose between bare-metal or virtual servers for their computing needs.

The IBM Cloud PaaS (Platform as a Service) offering is based on the open source cloud platform Cloud Foundry. This allows developers to use IBM services to create, manage, run and deploy various types of applications for the public cloud, in addition to local or on-premises environments. There's support for various programming languages, such as Java, Node.js, PHP and Python.

The IBM Cloud platform supports access to other IBM tools and services, such as IBM Watson and IBM Cloud Functions for serverless computing. The platform also supports access to tools and services from third-party vendors.

The IBM Cloud core offering consists of a full stack cloud platform with over 170 products and services covering data, containers, AI, IoT, and blockchain. IBM Cloud services are grouped into 16 categories, which include the following:

AI and Machine Learning

IBM Cloud users have the option to build their own machine learning models, or to use IBM Watson.

Subscribers who choose Watson have an ecosystem of tools and functions at their disposal. These include IBM Watson Assistant, which provides virtual agents that users can customize for any domain, and IBM Watson Discovery – a search and analytics engine that adapts to custom domains. The IBM Watson Knowledge Studio is a visual interface that teaches Watson domain-specific knowledge

The IBM Watson Knowledge Catalog provides a Software as a Service (SaaS) environment for AI data management, while the IBM Watson Studio is an Integrated Development Environment (IDE) for creating, running and managing AI models.

With generative AI and Natural Language Processing (NLP) now high on the agenda for many organisations, IBM Cloud's AI and machine learning services provide extensions for Watson in this area as well.

The IBM Watson Language Translator is an Application Programming Interface (API) for translation with domain-specific models, and the IBM Watson Natural Language Classifier provides a visual tool and API for text classification. IBM Watson Natural Language Understanding provides an API for text analysis and metadata extraction.

At the user level, there's IBM Watson Speech to Text – an API for real-time speech recognition and transcription. IBM Watson Text to Speech provides an API for real-time text to speech conversion.

Analytics

The IBM Cloud Analytics offering includes a range of data science tools, together with analytics services for streaming data.

The IBM Cloud Pak for Data offers tools for data analysis, organization, and management. IBM Analytics Engine is a Platform as a Service (PaaS) offering that enables users to build analytics applications on Apache Spark and Hadoop. IBM Master Data Management on Cloud provides managed master data storage, while IBM Streaming Analytics is a dashboard for real-time analysis of data streams.

IBM Cloud's Analytics offering also facilitates integration with other cloud environments. Notably, IBM Db2 Warehouse on Cloud offers a managed data warehouse on IBM Cloud or Amazon Web Services (AWS). IBM InfoSphere® Information Server on Cloud provides a data integration suite for Extract, Transform, Load (ETL), governance, and analysis.

Automation

Here, IBM Cloud Pak furnishes the principal suite of tools and services for automating business processes and workflows.

IBM Cloud Pak for Business Automation provides operations management software with AI insights, while IBM Cloud Pak for Integration is an AI-powered integration software solution for connecting apps and data across any cloud.

IBM Cloud Pak for Network Automation provides management software for operations on telecommunications networks, and the IBM Cloud Pak for Watson AIOps is a DevOps management tool with AI analysis and recommendations.

As an additional resource, IBM Cloud offers IBM Turbonomic® Application Resource Management, a software tool that automates application resource management and optimizes costs.

Blockchain

IBM's Blockchain Platform is a Software as a Service (SaaS) offering that enables subscribers to develop apps, enforce governance, and monitor a blockchain network. The IBM Cloud Blockchain Software Development Kit (SDK) is based on Hyperledger Fabric.

Compute

IBM Cloud offers a variety of compute resources for enterprises to host their workloads, including bare-metal servers, Virtual Machines (VMs), and serverless computing.

IBM Cloud Bare Metal Servers provide dedicated hardware for maximum computing performance, while the IBM Cloud for VMware Solutions offers hosting for virtualized workloads. IBM Cloud Virtual Servers for Classic Infrastructure offer the most flexible configuration options, and IBM Cloud Virtual Server for VPC provides virtual servers with enhanced network security.

For serverless environments, IBM Cloud Code Engine is a Platform as a Service offering for serverless container hosting.

In addition, IBM WebSphere Application Server on Cloud provides a runtime environment and Software Development Kit (SDK) for Java applications. For virtual desktop environments, Dizzion Managed DaaS on IBM Cloud offers a managed Desktop as a Service for virtual desktops in the cloud.

Containers

The IBM Cloud Kubernetes Service provides container hosting with self-healing and horizontal scaling. The platform gives user access to the Container Registry, a Software as a Service (SaaS) offering for container storage and management.

Istio provides a service mesh for microservices in Kubernetes clusters, while Red Hat® OpenShift® on IBM Cloud offers container hosting on managed Red Hat Linux servers.

Databases

For the storage and analysis of structured data, IBM Cloud offers a variety of SQL and NoSQL databases, as well as data querying, warehousing and migration tools.

Examples include IBM Cloud Databases for EnterpriseDB, a managed SQL database, and IBM Cloud Databases for MongoDB, a managed NoSQL JSON document store. IBM Informix® on Cloud offers a managed database for time series, spatial, NoSQL and SQL data.

Developer Tools

IBM Cloud CLI provides a Command Line Interface for developers to manage their IBM Cloud infrastructure. IBM Cloud Continuous Delivery provides a User Interface (UI) and CLI based DevOps workflows based on Tekton Pipelines. IBM Cloud App Configuration offers a Software Development Kit, dashboard and API for application feature management.

Other development tools on offer include IBM Cloud Messages for RabbitMQ, a managed message broker, IBM Cloud Schematics, a managed service to provision resources with terraform templates, and IBM Cloud Data Engine, a serverless, interactive querying service for analyzing object storage data.

Internet of Things (IoT)

The IBM IoT Platform provides services to connect and manage IoT devices, and tools for analysis of the data they produce.

For edge cloud network management (a key requirement for speedy and responsive IoT deployments), IBM Edge Application Manager provides an autonomous management platform for edge computing.

For IoT asset management, IBM Maximo Application Suite: Remote Monitoring enables system managers to see, predict and prevent issues, using advanced AI-powered remote monitoring and computer vision for assets and operations.

Quantum Computing

IBM Cloud's Quantum computing offering includes services and tools that enable organisations to run code on real quantum systems using a full-stack Software Development Kit (SDK). For example, Qiskit provides a Python SDK for open-source quantum development, while Qiskit Runtime offers a low-latency execution environment for quantum programs.

Security

The IBM Cloud Security portfolio includes services for activity tracking, identity and access management, and authentication. These include IBM Cloud App ID, a compliant authentication, authorization and user data SaaS, IBM Cloud Secrets Manager, a single-tenant, dedicated instance for managing secrets, and IBM Cloud Pak for Security, which provides tools for security auditing, reporting, analysis, and governance.

In addition to the above categories, IBM Cloud has infrastructure and service offerings that cover Networking, Logging and Monitoring, Integration, and Storage.

Pricing Considerations for IBM Cloud Users

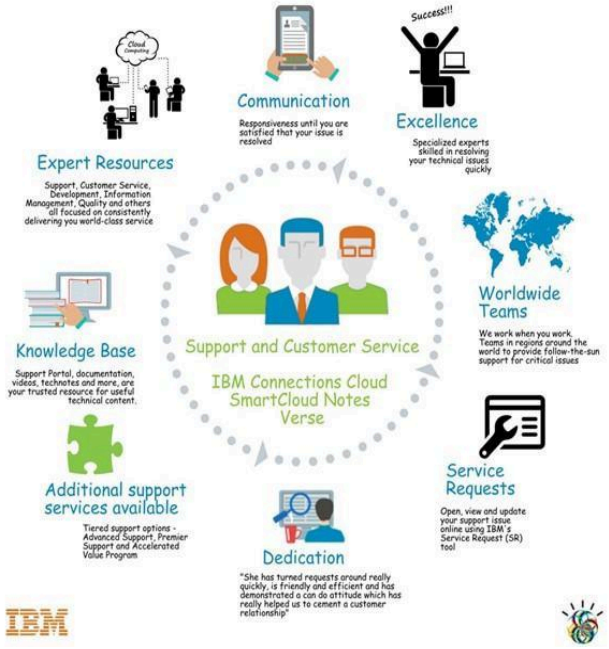
A number of factors will contribute to the actual cost of an IBM Cloud deployment, including resource usage, deployment model, and support levels. As of 2022, IBM Cloud has been offering two main pricing models.

IBM Cloud's pay-as-you-go (PAYG) accounts are billed monthly, based on resource consumption. By default, all new accounts are automatically created as pay-as-you-go plans.

For organisations willing to commit to a minimum amount of spending, the second pricing model is a subscription account. Here, larger monthly spending commitments earn bigger discounts.

There’s an IBM Cloud pricing calculator available, to help organisations estimate expenditure before committing to a particular type of plan.

How IBM Cloud is Evolving to Benefit Users



[Image source: CloudCompanions]

IBM Cloud is available in data centers worldwide, with coverage that extends across regions in North and South America, Europe, Asia, and Australia. Its next-generation hybrid cloud platform, deep enterprise expertise and advanced data and AI capabilities span 20 industries.

This combination of factors enables IBM Cloud to claim with some justification to offer the most open and secure public cloud for business. Its integrated offering combines Platform as a Service (PaaS) with Infrastructure as a Service (IaaS) to provide a solution that scales and supports both small development teams and organisations, and large scale enterprises.

IBM Cloud offers three deployment models.

1. **IBM Cloud Public** is a public cloud which provides access to virtual servers in a multi-tenant environment. Organisations can choose to deploy applications in one or multiple geographical regions.
2. **IBM Cloud Dedicated** is a single-tenant private cloud that IBM hosts in one of its data centers. Enterprises can connect to the environment using a direct network connection or VPN, leaving IBM to manage the platform.
3. **IBM Cloud Private** allows an organization to deploy the IBM platform as a private cloud in its own data center behind a firewall.

The Role of Linux in Cloud Computing and DevOps

Linux now cannot be underemphasized in the current and of the future era which includes the cloud computing and DevOps trend. It's open source character, flexibility and the ability to

scale account for the popularity of this cloud infrastructure tool by the organizations that intend to leverage it for software development and deployment process automation.

What Is Linux ?

Linux is an open-source operating system based on Unix. It was first introduced in 1991 by Linus Torvalds, a Finnish software engineer. Linux is known for its stability, security, and versatility, making it a popular choice for a wide range of applications, from personal computers to servers and supercomputers.

Features

Linux is renowned for its robust features that cater to the demands of cloud computing and DevOps:Linux is renowned for its robust features that cater to the demands of cloud computing and DevOps:

- **Open-source:** Linux's opens source character allows for the formulation of the community's development, the co-operation and frequent betterments.
- **Modularity:** Modular design of Linux posits users to rebuild the operating system troops o hert needs, avoiding unnecessary overhead and saving energy.
- **Security:** Since linux itself carries in its armor the security attributes like access controls and sandboxing, your critical operations are quite safe with it.
- **Compatibility:** Linux lives off the fact that it supports diverse hardware and software eliminating compatibility problems with the legacy environment.

Setting up Environment

Setting up a Linux environment for cloud computing and DevOps typically involves the following steps:

- **Choose a Linux distribution:** One of the favored Linux distributions among power users are Ubuntu, CentOS, and Debian mainly due to difference in features and level of support by the community.
- **Install virtualization software:** You can avoid the "dual boot" scenario and opt for tools such as VirtualBox or VMware to run Linux as a virtual machine on your local system.
- **Configure cloud instances:** Large cloud providers, for instance, Amazon Web Services, Google Cloud Platform and Microsoft Azure, provide Linux accessed through the virtual machines and container services that are easy to deploy.
- **Install DevOps tools:** Utilize the hardware such as Git, Jenkins, Docker, Kubernetes, and Ansible to your Linux work to help with the tasks of development and deployment processes.

Importance of linux

The importance of Linux lies in several factors:

- **Customizability:** Linux is much convenient, giving rise to the user's personalization options – introducing the unique behaviors to the operating system itself. Users can opt for something basic / a range of [Desktop Environments][window managers][software packages] according to their taste that eventually create a personalized environment.
- **Scalability:** Linux is very flexible able to be applied to a range of tasks, from embedded systems (systems that execute programs entirely or excluding external processors) to

high-performance computing clusters. It is modular by nature thus scaling and changing it into the software and hardware requirements comes easily.

- **Cost-Effectiveness:** However, free to use, distribute, and make changes in the software Linux would have a competitive solution in specific areas like educational institutions, businesses, and organizations. This explains its reasons for massive adoption which is an additional advantage to it.
- **Community Support:** Linux has a community of programmers, those that actually use the system, and enthusiasts who form a team of developers that support, share the experience, and the skills with one another.

The Role of Linux in Cloud Computing

Linux is the main contributor in the modern cloud via using Linux based infrastructures for creating many cloud platforms and services Here are some key points about the role of Linux in cloud computing:The following are important themes concerning the Linux in Cloud Computing: Browse through our curated collection of mind-provoking questions to expand your understanding of familiar topics and broaden your intellectual horizons.

- **Virtualization:** Linux is a great choice to use as an operating system for VMs which are required in cloud environments for both creating and managing them.
- **Cloud Infrastructure:** Main cloud carriers, in particular Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform, have started using Linux as an OS on the bottom tier of their cloud infrastructure.
- **Container Technologies:** Linux functioned as a base for the container technologies, such as Docker and Kubernetes, which now are widely adopted for deployment and administration of applications on cloud environments.
- **Scalability and Flexibility:** The scalable and customizable qualities of Linux mean that it's a good fit for cloud atmospheres where resources need to be reallocated and scaled accordingly to match demand at any given time.
- **Open-Source Ecosystem:** The Linux open source architecture and the ability to customize the operating system and tools with many cloud service providers and developers roles, that Linux provides an environment for its integration with various varieties.

Advantages of Linux in Cloud Computing

- **Cost-Effective:** Linux is an open source and is therefore free, thus maintaining its open infrastructure and services at much lower costs.
- **Scalability:** Scalability of Linux provides cloud providers with a straightforward upgrade or decrease opportunity in the amount of infrastructure as the demand increases or drops.
- **Security:** Platform to Cloud Security is a well-known area for Linux, which ensures that the operating system platform is a secure platform for a cloud.
- **Compatibility:** One of the great things about Linux operating system is that it supports most of the hardware and software which you may need for the cloud based services in your applications, to ensure smooth working of the whole system.
- **Community Support:** The huge Linux community showcases timely support, updates, and contributions to improve, modify and upgrade the applications to achieve perfection.

Disadvantages of Linux in Cloud Computing

- **Compatibility Issues:** Through the fact that Linux is usually compatible with most presented software and hardware, some incompatibility situations with specific proprietary applications or drivers still could arise.
- **Steep Learning Curve:** Linux has a greater (tougher) learning curve compared to some proprietary (commercial) operating systems that may be problematic for users and admins who instead for example are not familiar with its command-line interface and configuration.
- **Limited Desktop Applications:** Linux can boast about the vast range of server and command-line applications, however, the desktop apps could be scarce and need to be installed manually when compared to other systems.
- **Driver Support:** The lack of official support for hardware drivers by some hardware manufacturers is arguably the biggest problem with creating a compatible Linux system as it can limit the hardware compatibility or requires the user to use community-developed drivers.

The Role of Linux in DevOps

Linux speaks out a large role in DevOps (Development and Operations) ideology which aims to mend the gaps of the software developers and IT administrators. Here are some key points about the role of Linux in DevOps:Here are some key points about the role of Linux in DevOps:

- **Containerization:** The bottom-line for containerize solutions like Docker and Kubernetes which are the most used in DevOps for the application packaging, deployment, and management lies in the fact that there are based on Linux.
- **Automation and Configuration Management:** Operating system Linux gives a divergent choice of apparatuses and utilities to address automating tasks and managing configurations, as Ansible, Puppet and Chef are indeed indispensable to approach DevOps.
- **Continuous Integration and Continuous Deployment (CI/CD):** In so many cases, mainstream continuous integration and continuous deployment tools like Jenkins, Gitlab CI/CD and Travis CI are popular and over widely used in DevOps workflow pipelines.
- **Cloud-Native Development:** Cloud native development relies on a Linux kernel as being the core operating system component, a fact that is fundamental to DevOps, as it allows engineers to build and deploy applications that are designed to run on such cloud environments.
- **Scripting and Automation:** One the areas where linux excels is its command-line interface and scripting capabilities (e.g., Bash Python). These make automating of various tasks and processes like DevOps practices easy.

Advantages of Linux in DevOps

- **Open-Source and Customizable:** Open-source architecture and extensibility of Linux allow for seamless linkage with multiple DevOps as well as overall workflow tools.
- **Stability and Reliability:** Linux has a reputation for its stability and reliability such that it is a huge selling point in solving workflow problems that require a high level of security in DevOps environments.
- **Scalability:** Scalable Linux makes it possible for DevOps teams to threshold growth barriers in their infrastructure and apps without limitations.

- **Community Support:** The big Linux community delivers the tools and resources even the support required for the DevOps practices.
- **Cost-Effective:** Open source and free-of-charge implementation of Linux decreases the DevOps operation's costs in general.

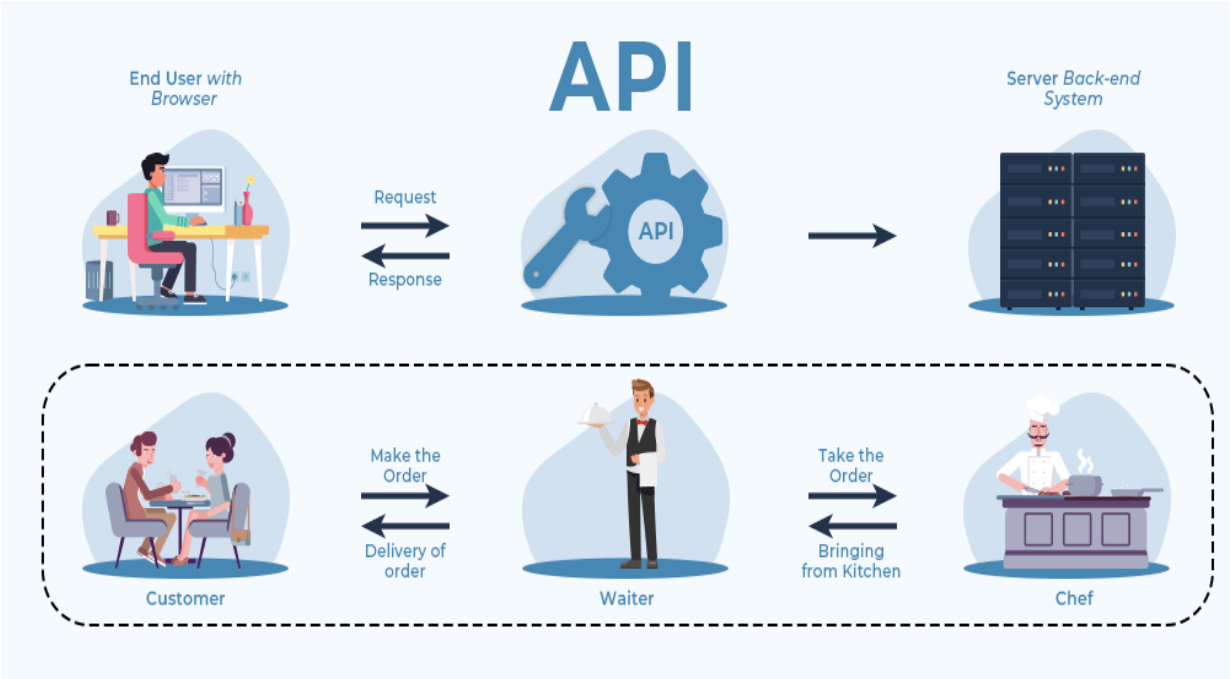
Disadvantages of Linux in DevOps

- **Learning Curve:** While Linux is less intuitive than some proprietary operating systems, whose use is familiar to many individuals and teams, its command-line interface and configuration exercises a steeper learning curve for those who have yet to be exposed to it before.
- **Limited GUI Tools:** Linux utilizes a robust command line tool set for devops work. However, GUI Tools may be harder to discover than other operating systems are.
- **Hardware Compatibility Issues:** Equipment makers sometimes do not deliver well-developed Linux managedivews which in some cases cause software-hardware compatibility or in worst cases when alternative options from the community is involved.
- **Vendor Support:** Although Linux has an immense community, some enterprises can go for the proven reliability of vendor-supported solutions instead of community-based open source products as this option lets organizations have guaranteed support and security of their DevOps infrastructure.

API (Application Programming Interface) :

Everyone is in search of the highest-paying job so as to get into it. And, in the list, the web developer has been on the top for years and will remain in the same place due to its demand. If you're the one who's looking to get into it. you must be aware of the most important terms used in it. Out of all the terms, API is yet another term that plays a very important role in building a website. Now,

What is an API – (Application Programming Interface)?



To make you clear with the diagram of **what is API** , let's take **a real-life example of an API,** you can think of an API as a waiter in a restaurant who listens to your order request, goes to the chef, takes the food items ordered and gets back to you with the order. Also, if you want to look for the working of an API with the example, here's one. *You're searching for a course* ***(let's say DSA-Self Paced)*** *on the* ***XYZ*** *website, you send a request(product search requested) through an API, and the database searches for the course and checks if it's available, the* ***API*** *is responsible here to send your request to the database (in search of the course) and responds with the output(best DSA courses).*

What is an API?

API full form is an **Application Programming Interface** that is a collection of communication protocols and subroutines used by various programs to communicate between them. A programmer can make use of various API tools to make their program easier and simpler. Also, an API facilitates programmers with an efficient way to develop their software programs. Thus **api meaning** is when an API helps two programs or applications to communicate with each other by providing them with the necessary tools and functions. It takes the request from the user and sends it to the service provider and then again sends the result generated from the service provider to the desired user.

A developer extensively uses APIs in his software to implement various features by using an API call without writing complex codes for the same. We can create an API for an **operating system** , **database system** , **hardware system, JavaScript file** , or similar object-oriented files. Also, an API is similar to a GUI(Graphical User Interface) with one major difference. Unlike GUIs, an **application program interface** helps software developers to access web tools while a GUI helps to make a program easier to understand for users.

APIs are the building blocks for the todays websites in which heavy data is transferred from the client to server and vice versa. If you want to learn such more concepts of the websites then you should enrol in our Full Stack Node Development Course

How do APIs Work?

The working of an API can be clearly explained with a few simple steps. Think of a client-server architecture where the client sends the request via a medium to the server and receives the response through the same medium. An API acts as a communication medium between two programs or systems for functioning. The client is the user/customer (who sends the request), the medium is the **application interface programming** , and the server is the backend (where the request is accepted and a response is provided). Steps followed in the working of APIs –

- The client initiates the requests via the APIs URI (Uniform Resource Identifier)
- The API makes a call to the server after receiving the request
- Then the server sends the response back to the API with the information
- Finally, the API transfers the data to the client

APIs are considered safe in terms of attacks as it includes authorization credentials and an API gateway to limit access so as to minimize security threats. To provide additional security layers to the data, HTTP headers, query string parameters, or cookies are used.

If we talk about the architectures, API's architectures are:

- **REST (Representational State Transfer)**
- **SOAP (Simple Object Access Protocol)**

Both define a standard communication protocol for the exchange of messages in **XML** (Extensible Markup Language).

How is an API Different From a Web Application?

An API acts as an interface that allows proper communication between two programs whereas a web application is a network-based resource responsible for completing a single task. Also, it's important to know that **"All web services are APIs, but not all APIs are web".**

The difference between an API and a web application is that API allows two-way communication and web applications are just a way for users to interact through a web browser. A web application may have an API to complete the requests.

Types of APIs

There are three basic forms of API –

1. WEB APIs

A **Web API** also called Web Services is an extensively used API over the web and can be easily accessed using the HTTP protocols. A Web **application programming interface is** an open-source interface and can be used by a large number of clients through their phones, tablets, or PCs.

2. LOCAL APIs

In this type of API, the programmers get the local middleware services. TAPI (Telephony Application Programming Interface), and .NET are common examples of Local APIs.

3. PROGRAM APIs

It makes a remote program appear to be local by making use of RPCs (Remote Procedural Calls). SOAP is a well-known example of this type of API.

Few other types of APIs:

- **SOAP (SIMPLE OBJECT ACCESS PROTOCOL):** It defines messages in XML format used by web applications to communicate with each other.
- **REST (Representational State Transfer):** It makes use of HTTP to GET, POST, PUT, or DELETE data. It is basically used to take advantage of the existing data.
- **JSON-RPC:** It uses JSON for data transfer and is a lightweight remote procedural call defining a few data structure types.
- **XML-RPC:** It is based on XML and uses HTTP for data transfer. This API is widely used to exchange information between two or more networks.

What are REST APIs?

REST stands for Representational State Transfer, and follows the constraints of REST architecture allowing interaction with RESTful web services. It defines a set of functions (GET, PUT, POST, DELETE) that clients use to access server data. The functions used are:

- GET (retrieve a record)
- PUT (update a record)
- POST (create a record)
- DELETE (delete the record)

Its main feature is that REST API is stateless, i.e., the servers do not save clients' data between requests.

is a Web API?

Web API Is simply an API for the web. It is an API that can be accessed using the HTTP protocol. It can be built using Java, .nET, etc. It is implemented to extend the functionality of a browser, simplify complex functions, and provide easy syntax to complex code.

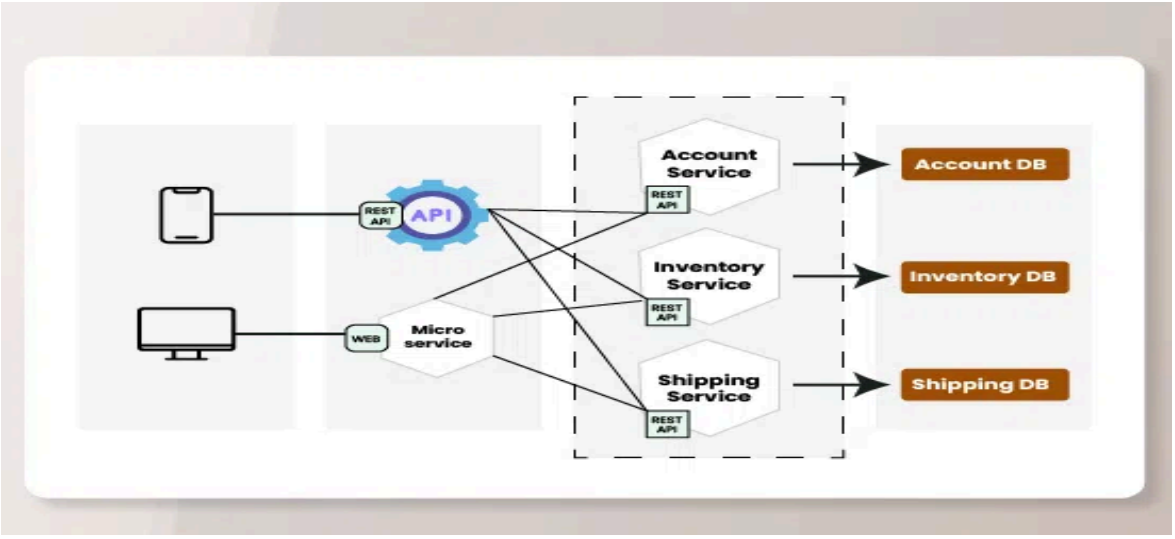
The four main types of web APIs are:

- Open API
- Partner API
- Internal API
- Composite API

Microservices:

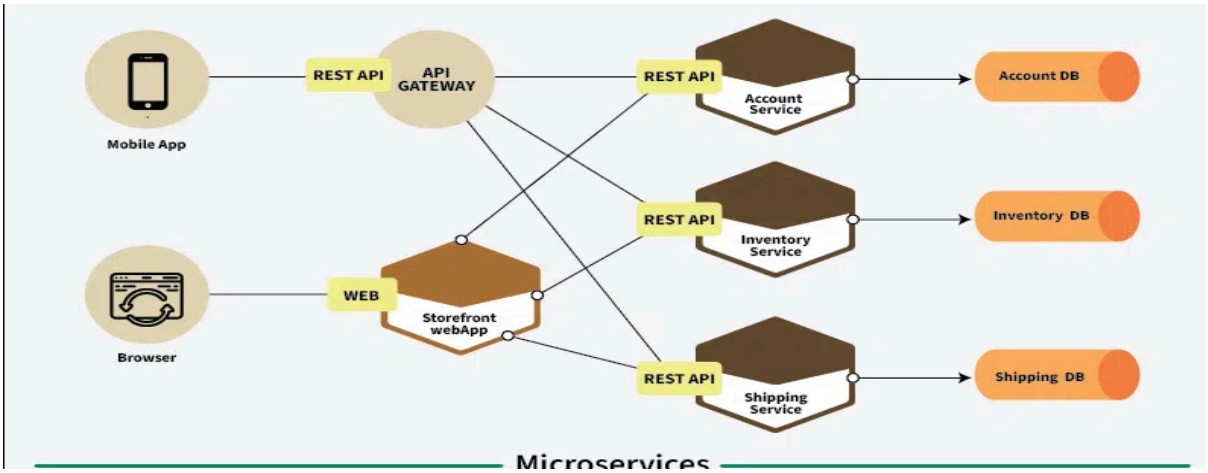
Microservices are an architectural approach to developing software applications as a collection of small, independent services that communicate with each other over a network. Instead of building a monolithic application where all the functionality is tightly integrated

into a single codebase, microservices break down the application into smaller, loosely coupled services.



What are Microservices?

Microservice is a small, loosely coupled service that is designed to perform a specific business function and each microservice can be developed, deployed, and scaled independently.



- This architecture allow you to take a large monolith application and decompose it into small manageable components/services. Also, it is considered as the building block of modern applications.
- Microservices can be written in a variety of programming languages, and frameworks, and each service acts as a mini-application on its own.

How do Microservices work?

Microservices break complex applications into smaller, independent services that work together, enhancing scalability, and maintenance. Below is how microservixes work:

- Applications are divided into self-contained services, each focused on a specific function, simplifying development and maintenance.
- Each microservice handles a particular business feature, like user authentication or product management, allowing for specialized development.
- Services interact via APIs, facilitating standardized information exchange and integration.
- Different technologies can be used for each service, enabling teams to select the best tools for their needs.
- Microservices can be updated independently, reducing risks during changes and enhancing system resilience.

What are the main components of Microservices Architecture?

Main components of microservices architecture include:

- **Microservices:** Small, loosely coupled services that handle specific business functions, each focusing on a distinct capability.
- **API Gateway:** Acts as a central entry point for external clients also they manage requests, authentication and route the requests to the appropriate microservice.
- **Service Registry and Discovery:** Keeps track of the locations and addresses of all microservices, enabling them to locate and communicate with each other dynamically.
- **Load Balancer:** Distributes incoming traffic across multiple service instances and prevent any of the microservice from being overwhelmed.
- **Containerization:** Docker encapsulate microservices and their dependencies and orchestration tools like Kubernetes manage their deployment and scaling.
- **Event Bus/Message Broker:** Facilitates communication between microservices, allowing pub/sub asynchronous interaction of events between omponents/microservices.
- **Database per Microservice:** Each microservice usually has its own database, promoting data autonomy and allowing for independent management and scaling.
- **Caching:** Cache stores frequently accessed data close to the microservice which improved performance by reducing the repetitive queries.
- **Fault Tolerance and Resilience Components:** Components like circuit breakers and retry mechanisms ensure that the system can handle failures gracefully, maintaining overall functionality.

Design Patterns for Microservices Architecture

Below are the main design pattern of microservices:

1. API Gateway Pattern

Think of the API gateway as the front door to your microservices. It acts as a single point of entry for clients, managing requests and directing them to the appropriate service.

This pattern simplifies the client's experience by hiding the complexities of multiple services behind one interface. It can also handle tasks like authentication, logging, and rate limiting, making it a crucial part of microservices architecture.

2. Service Registry Pattern

This pattern is like a phone book for microservices. It maintains a list of all active services and their locations (network addresses). When a service starts, it registers itself with the registry.

Other services can then look up the registry to find and communicate with it. This dynamic discovery enables flexibility and helps services interact without hardcoding their locations.

3. Circuit Breaker Pattern

Imagine a circuit breaker in your home that stops electricity flow when there's an overload. Similarly, this pattern helps protect your system from cascading failures.

If a service fails repeatedly, the circuit breaker trips, preventing further requests to that service. After a timeout period, it allows limited requests to test if the service is back online. This reduces the load on failing services and enhances system resilience.

4. Saga Pattern

This pattern is useful for managing complex business processes that span multiple services. Instead of treating the process as a single transaction, the saga breaks it down into smaller steps, each handled by different services.

If one step fails, compensating actions are taken to reverse the previous steps. This way, you maintain data consistency across the system, even in the face of failures.

5. Event Sourcing Pattern

Instead of storing just the current state of an application, this pattern records all changes as a sequence of events.

Each event describes a change that occurred, allowing services to reconstruct the current state by replaying the event history. This provides a clear audit trail and simplifies data recovery in case of errors.

6. Strangler Fig Pattern

This pattern allows for a gradual transition from a monolithic application to microservices. New features are developed as microservices while the old system remains in use.

Over time, as more functionality is moved to microservices, the old system is gradually "strangled" until it can be fully retired. This approach minimizes risk and allows for a smoother migration.

7. Bulkhead Pattern

Similar to compartments in a ship, the bulkhead pattern isolates different services to prevent failures from affecting the entire system.

If one service encounters an issue, it won't compromise others. By creating boundaries, this pattern enhances the resilience of the system, ensuring that a failure in one area doesn't lead to a total system breakdown.

8. API Composition Pattern

When you need to gather data from multiple microservices, the API composition pattern helps you do so efficiently.

A separate service (the composition service) collects responses from various services and combines them into a single response for the client. This reduces the need for clients to make multiple requests and simplifies their interaction with the system.

9. CQRS Design Pattern

CQRS divides the way data is handled into two parts: commands and queries. Commands are used to change data, like creating or updating records, while queries are used just to fetch data. This separation allows you to tailor each part for its specific purpose.

For instance, the command side can focus on enforcing business rules, while the query side can be optimized for fast data retrieval. This pattern is especially helpful in applications with a lot of read and write operations, as it enhances performance and scalability by allowing for different optimizations for each side.

Anti-Patterns for Microservices Architeture

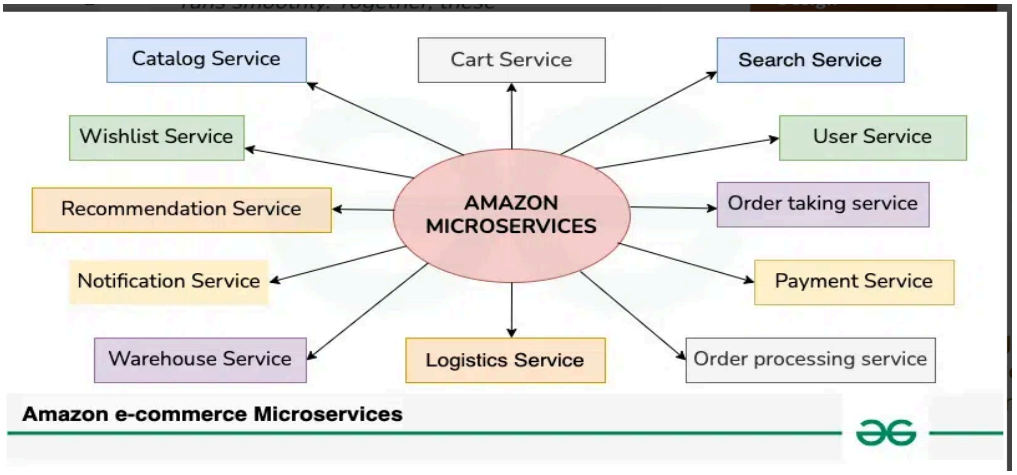
Learning anti-patterns in microservices is crucial for avoiding common mistakes. Below are some anti-patterns in microservices and by understanding these anti-patterns, developers can make informed decisions and implement best practices.

- When microservices share a single centralized database, it can compromise their independence and scalability.
- Microservices that frequently communicate for minor tasks can create excessive network traffic, leading to delays and increased latency.
- Creating too many microservices for small functions can add unnecessary complexity to the system.
- If the boundaries between microservices are not clearly defined, it can cause confusion about their responsibilities.
- Failing to address security issues in microservices can expose the system to vulnerabilities and potential data breaches.

Real-World Example of Microservices

Let's understand the Miscroservices using the real-world example of Amazon E-Commerce Application:

Amazon's online store is like a giant puzzle made of many small, specialized pieces called microservices. Each microservice does a specific job to make sure everything runs smoothly. Together, these microservices work behind the scenes to give you a great shopping experience.



Below are the microservices involved in Amazon E-commerce Application:

- **User Service**: Handles user accounts and preferences, making sure each person has a personalized experience.
- **Search Service**: Helps users find products quickly by organizing and indexing product information.
- **Catalog Service**: Manages the product listings, ensuring all details are accurate and easy to access.
- **Cart Service**: Lets users add, remove, or change items in their shopping cart before checking out.
- **Wishlist Service**: Allows users to save items for later, helping them keep track of products they want.

- **Order Taking Service**: Processes customer orders, checking availability and validating details.
- **Order Processing Service**: Oversees the entire fulfillment process, working with inventory and shipping to get orders delivered.
- **Payment Service**: Manages secure transactions and keeps track of payment details.
- **Logistics Service**: Coordinates everything related to delivery, including shipping costs and tracking.
- **Warehouse Service**: Keeps an eye on inventory levels and helps with restocking when needed.
- **Notification Service**: Sends updates to users about their orders and any special offers.
- **Recommendation Service**: Suggests products to users based on their browsing and purchase history

Microservices vs. Monolithic Architecture

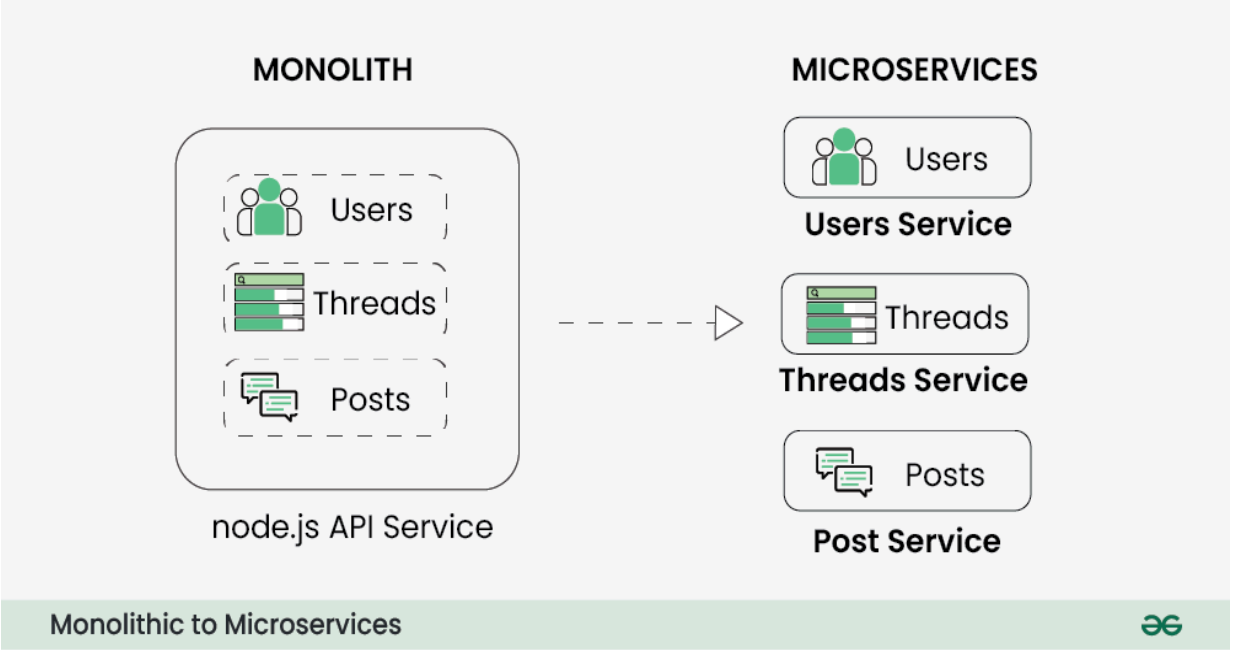
Below is a tabular comparison between microservices and monolithic architecture across various aspects:

Aspect	**Microservices Architecture**	**Monolithic Architecture**
Architecture Style	Decomposed into small, independent services.	Single, tightly integrated codebase.
Development Team Structure	Small, cross-functional teams for each microservice.	Larger, centralized development team.
Scalability	Independent scaling of individual services.	Scaling involves replicating the entire application.
Deployment	Independent deployment of services.	Whole application is deployed as a single unit.
Resource Utilization	Efficient use of resources as services can scale independently.	Resources allocated based on the overall application's needs.
Development Speed	Faster development and deployment cycles.	Slower development and deployment due to the entire codebase.

Aspect	Microservices Architecture	Monolithic Architecture
Flexibility	Easier to adopt new technologies for specific services.	Limited flexibility due to a common technology stack.
Maintenance	Easier maintenance of smaller, focused codebases.	Maintenance can be complex for a large, monolithic codebase.

How to migrate from Monolithic to Microservices Architecture?

Below are the main the key steps to migrate from a monolithic to microservices architecture:



- **Step 1**: Begin by evaluating your current monolithic application. Identify its components and determine which parts can be transitioned into microservices.
- **Step 2**: Break down the monolith into specific business functions. Each microservice should represent a distinct capability that aligns with your business needs.
- **Step 3**: Implement the Strangler Pattern to gradually replace parts of the monolithic application with microservices. This method allows for a smooth migration without a complete overhaul at once.
- **Step 4**: Establish clear APIs and contracts for your microservices. This ensures they can communicate effectively and interact seamlessly.
- **Step 5**: Create Continuous Integration and Continuous Deployment (CI/CD) pipelines. This automates testing and deployment, enabling faster and more reliable releases.
- **Step 6**: Introduce mechanisms for service discovery so that microservices can dynamically locate and communicate with each other, enhancing flexibility.

- **Step 7**: Set up centralized logging and monitoring tools. This provides insights into the performance of your microservices, helping to identify and resolve issues quickly.
- **Step 8**: Ensure consistent management of cross-cutting concerns, such as security and authentication, across all microservices to maintain system integrity.
- **Step 9**: Take an iterative approach to your microservices architecture. Continuously refine and expand your services based on feedback and changing requirements

Service-Oriented Architecture(SOA) vs. Microservices Architecture

Below is a tabular comparison between Service-Oriented Architecture (SOA) and Microservices across various aspects:

Aspect	**Service-Oriented Architecture(SOA)**	**Microservices Architecture**
Scope	Includes a broad set of architectural principles.	Focuses on building small, independent services.
Size of Services	Services tend to be larger and more comprehensive.	Services are small, focused, and single-purpose.
Data Management	Common data model and shared databases are common.	Each service has its own database or data store.
Communication	Typically relies on standardized protocols like SOAP.	Uses lightweight protocols such as REST or messaging.
Technology Diversity	Can have different technologies, but often standardized middleware.	Encourages diverse technologies for each service.
Deployment	Services are often deployed independently.	Promotes independent deployment of microservices.
Scalability	Horizontal scaling of entire services is common.	Enables independent scaling of individual services.
Development Speed	Slower development cycles due to larger services.	Faster development cycles with smaller services.

Aspect	Service-Oriented Architecture(SOA)	Microservices Architecture
Flexibility	Can be flexible, but changes may affect multiple services.	Provides flexibility due to independent services.

Benefits and Challenges of using Microservices Architecture

Benefits of Using Microservices Architecture:

- Teams can work on different microservices simultaneously.
- Issues in one service do not impact others, enhancing reliability.
- Each service can be scaled based on its specific needs.
- The system can quickly adapt to changing workloads.
- Teams can choose the best tech stack for each microservice.
- Small, cross-functional teams work independently.

Challenges of using Microservices Architecture

- Managing service communication, network latency, and data consistency can be difficult.
- Decomposing an app into microservices adds complexity in development, testing, and deployment.
- Network communication can lead to higher latency and complicates error handling.
- Maintaining consistent data across services is challenging, and distributed transactions can be complex.

Real-World Examples of Companies using Microservices Architecture

Organizations have undergone significant changes by adopting microservices, moving from monolithic applications. Here are some real-life examples:

- **Amazon:** Initially a monolithic app, Amazon uses microservices early on, breaking its platform into smaller components. This shift allowed for individual feature updates, greatly enhancing functionality.
- **Netflix:** After facing service outages while transitioning to a movie-streaming service in 2007, Netflix adopted a microservices architecture. This change improved reliability and performance.
- **Uber:** By switching from a monolithic structure to microservices, Uber operations were become smoother, resulting in increased webpage views and search efficiency

Docker:

Docker is a powerful tool used for developing, packaging, and deploying applications efficiently. Docker is a container management service. Docker was released in 2013. It is open-source and available for different platforms like Windows, macOS, and Linux. Docker is quickly shipping, testing, and deploying code.

So that it reduces your delay between writing code and running it in production. You can create self-contained environments known as containers. That can run consistently on different platforms.
You need to install the Docker engine on your computer or device. The Docker engine allows you to create and manage docker containers, docker images, docker hub, docker desktop, etc.

Docker Installation
This section will walk you through the steps of installing Docker on your system, whether it's Windows, macOS, or Linux. We'll cover the different installation methods, system requirements, and post-installation configurations to ensure you have Docker up and running smoothly, ready to build and run your containerized applications.

- Docker Installation on Windows
- Docker Installation on Windows using Chocolatey
- Docker Installation on Ubuntu
- Docker Installation on macOS
- Docker Installation on Arch-based Linux
- Docker Installation on RedHat

Docker Commands
In this part of the Docker tutorial we will dives into the essential commands you need to navigate the world of Docker. Whether you're building, running, or managing containers, this guide equips you with the tools to control your containerized applications.

- Docker Instructions
- Run Commands Inside Containers
- USER Instruction

Docker Run
Docker Run refers to the command used in Docker to create and start containers based on Docker images. It's a fundamental aspect of working with Docker, allowing users to deploy applications and services quickly and efficiently within isolated environments known as containers. Here in this section, you will get to know all the details about the Docker Run.

- Dockerfile
- Syntax of Dockerfile
- How to Build a Web Server Docker File?

Docker Images
A Docker Image contains everything a container needs to run, including the application code, libraries, dependencies, and the operating system it needs.

- Docker Images
- Working with Images
- Publish Docker Images in Hub
- Create a Customized Image
- Use Image Tags
- How to use Next.js Image
- How to use Local Image with Minikube

Docker Compose

Managing multiple containers can get complex! Docker Compose simplifies this process. In this section we have listed down all the details about the Docker Compose like intro and Compose tools.

- Introduction to Docker Compose
- Compose tools to run Multi Container Apps

Docker Engine, Storage

Docker Engine, also known as Docker Daemon, is the core component of the Docker platform responsible for running and managing Docker containers. Explore this section to get all details about Docker Engine.

- Docker Storage
- Docker Data Storage
- Backup Docker Container
- Manage Volumes using CLI

Docker Networking

Docker Networking refers to the set of mechanisms and technologies Docker provides for communication between Docker containers, as well as between containers and the outside world. Go through this section to get more details about Docker Networking.

- Docker Networking
- Docker Ports
- Creating a Network and connecting a Container
- Connecting Two Docker Containers Over the Same Network
- Default Bridge Networking
- Create your own secure Home Network using Pi-hole and Docker

Docker Registry

After knowing the Docker Networking, in Docker tutorial, in this section we are going to discuss Docker Registry which is refer as central repository for storing the and managing Docker image.

- Docker Registry
- Docker – Public Repositories
- Docker – Private Registries
- Creating a Private Repository and Push an Image to That Private Repository
- Using Public Repositories To Host Docker Images

Docker Containers and Managing Containers

Explores this section to get to know about the fundamental concepts and practical aspects of utilizing Docker technology to deploy and manage software applications efficiently in this section.

- Containerization using Docker
- Docker Container Virtualization
- Docker Container for Node.js

- Remove Containers
- Push a Container Image to a Docker Repository
- Docker Container Linking
- Manage Containers using CLI
- Mount Volume inside Container
- Difference between Images and Container
- Difference between Virtual Machines and Containers
- How to Install Linux Package in Docker Container
- Copying Files to and from Docker Containers
- Run MongoDB as Container
- Container for Node.js
- Container for NGNIX
- How to Provide the Static IP to a Docker Container?

Docker Swarm

Docker Swarm steps in as your reliable organizer, making sure all your containers are in the right place and working well together. Go through this section to get an all about Docker Swarm.

- Introduction to Docker Swarm
- Difference between Kubernetes and Docker Swarm

Miscellaneous

- Running a Java Application using Docker
- Running a PHP Application using Docker
- Running a Python Application using Docker
- How to Dockerize an ExpressJS App?
- How to Dockerize a ReactJS App?
- Docker Vs Kubernetes
- Deploying WebApps on Docker
- Docker Continuous Integration
- Difference Between Vagrant and Docker
- How to Setup Jenkins in Docker Container?

Docker Container: A Docker container is a lightweight and executable package of software. It includes everything used to run an application, code, libraries, and dependencies. You don't need to allocate any memory for the application. It can automatically generate space according to the requirements.

Why to learn Docker?

A couple of years back, when organizations needed other applications, they buy a server without knowing the performance requirement of the software/application. This results waste of money and resources. Then Virtual machines come onto the market, which allows engineers to run multiple applications on the same resource but as a completely different server. But Every application to run on a Virtual machine requires OS and every OS needs its own CPU, RAM, etc. to run. Which increases the cost.

Then Docker Model comes into the scope, which overcomes the drawbacks of the Virtual Machine. It reduces the wastage of resources by sharing OS, memory, and CPU, and It offers many benefits for developers and system administrators, like consistency, portability, efficiency, security, scalability, and version control. These benefits make it easier to manage and deploy applications.

Features of Docker

- Docker reduces the size of development by providing a smaller part of the OS via containers.
- It is easier to work on the same project by different teams with the help of Containers.
- Docker containers can be deployed anywhere, on any physical, or virtual machines and on the cloud.
- Docker containers are lightweight so, it becomes easy to scale them.

Advantages of Docker

- Docker uses less memory.
- The full operating system is not required to run an application.
- Containers run faster than the other Virtual Machines.
- It is lightweight.
- It allows us to use a remote repository to share your containers with other teams.
- To reduce the risks, it uses dependencies.

Disadvantages of Docker

- Complexity will increase due to layering.
- It is difficult to manage a large number of containers.
- For an application that needs better graphics, Docker is not suitable for it.
- Cross-platform compatibility is not allowed.

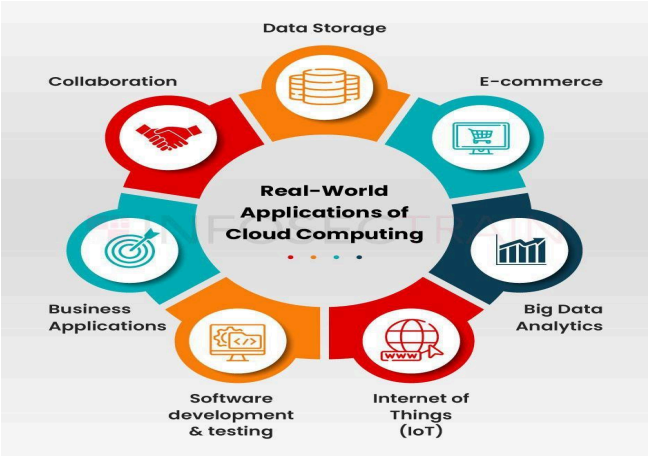
Real-world applications of cloud computing

Cloud technology has many applications across industries, including business, data storage, management, entertainment, social networking, education, and art. There are three primary types of cloud computing service models: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS).

Several cloud platforms are available, each with its own set of features and benefits, such as Public Cloud, Private Cloud, Hybrid Cloud, and Community Cloud.

Cloud computing has numerous applications in a variety of industries.

Some examples of real-world cloud computing applications include:



1. Data storage

Cloud storage provides individuals and businesses with a reliable and cost-effective way to store and access data remotely. Cloud storage providers offer different security, backup, and accessibility options, making it easier for users to manage and protect their data. With cloud storage, users can access their files and data from any device with an Internet connection, eliminating the need for physical storage devices and making it easier to share and collaborate on files with others.

2. Software development and testing

Cloud platforms offer developers an easy way to create, test, and deploy applications without expensive infrastructure. Cloud development platforms provide a secure and scalable environment for software development, allowing developers to build and deploy applications more quickly and efficiently. With cloud computing, developers can access the necessary tools and resources, such as Software Development Kits (SDKs), testing tools, and virtual machines, to create and test their software applications.

3. Business applications

Cloud-based business applications, such as Customer Relationship Management (CRM) and Enterprise Resource Planning (ERP), provide businesses with a scalable and cost-effective way to manage their operations. Cloud-based business applications are accessible from any device with Internet connectivity, making it simpler for employees to collaborate and work remotely. Cloud-based business applications also provide businesses with real-time data insights and analytics, enabling better decision-making and a competitive edge.

4. Collaboration

Cloud-based collaboration tools, such as Google Docs and Microsoft Office 365, allow teams

to work together on documents, presentations, and other projects in real time from anywhere with an Internet connection. Cloud-based collaboration tools offer a secure and efficient way to collaborate on projects, allowing team members to make changes and track revisions in real time. Cloud-based collaboration tools also provide businesses with better version control and audit trails, making tracking changes and managing workflows easier.

5. Big data analytics

Big data analytics platforms hosted in the cloud enable businesses to process and analyze information swiftly and efficiently, resulting in improved insights and decision-making. Cloud-based big data analytics platforms provide a scalable and cost-effective method for storing, processing, and analyzing data, making it easier for businesses to manage and extract value from their data. With cloud-based big data analytics, companies can quickly identify patterns, trends, and insights, leading to better business decisions and improved performance.

6. E-commerce

Cloud-based e-commerce platforms provide businesses with a scalable and reliable way to manage online transactions, inventory, and customer data. Cloud-based e-commerce platforms offer companies the flexibility and scalability to handle increased traffic and demand, ensuring customers have a seamless shopping experience. With cloud-based e-commerce, companies can easily manage their online stores, inventory, and payments while providing customers with a secure and efficient way to shop.

7. Internet of Things (IoT)

Cloud platforms can provide a centralized hub for managing IoT devices and data, enabling businesses to develop and deploy IoT solutions more efficiently and cost-effectively. With cloud-based IoT, companies can collect, store, and analyze data from multiple IoT devices in real-time, leading to better insights and decision-making. It also provides businesses with better security and scalability options, making it easier to manage and scale IoT solutions as needed.

Cloud computing has various uses across various industries. It provides businesses and individuals with a reliable, scalable, cost-effective way to manage their operations, store and access data, collaborate on projects, and develop and deploy applications.